Unit-1:Introduction to contemporary technology

Contents

Introduction

Contemporary technology refers to the belonging to the same age, living or occurring in the same period of time existing or occurring at the present time, conforming to modern or current ideas in cloud computing, artificial intelligence, Virtual Reality e-governance, mobile computing and Internet of things.

The Importance of Modern Technology

Modern technology has changed civilization in many different ways. Humans have almost always been on a path of progression, but thanks to technology, the twentieth and twenty-first centuries have seen a number of advancements that revolutionized the way people work, live and play. Imagining what life would be like without some of these advancements has become a difficult task due to their importance and our reliance on them.

❖ **Communication**

One of the areas where modern technology is most important is in the realm of communication. Long ago, communicating with people outside your immediate location was a difficult process, requiring communication by physical letter and a lot of patience. The Internet has made long distances almost transparent, allowing users to correspond with people on the other side of the planet in an instant. Technology has also increased our connectivity, with cell phones and other devices providing an always-on link to the global communication network.

❖ **Education**

Another area where computers and the Internet have become extremely important is in education. Computers can store large amounts of data in a very small space, reducing entire shelves of reference books down to a single CD of data. The Internet also serves as a massive resource for learning, linking informational sites together and allowing the curious to search for almost any topic imaginable. A single computer could store hundreds of educational games, audio and visual lessons as well as provide access to a wealth of knowledge for students. In the classroom, virtual whiteboards can replace blackboards, allowing teachers to provide interactive content for students and play educational films without the need to set up a projector.

❖ **Health**

Technology has also had significant effects on the health care industry. Advancements in diagnostic tools allow doctors to identify diseases and conditions early, increasing the chances of a successful treatment and saving lives. Advancements in medicines and vaccines have also proven extremely effective, nearly eradicating diseases like measles, diphtheria and smallpox that once caused massive epidemics. Modern medicine also allows patients to manage chronic conditions that were once debilitating and life-threatening, such as diabetes and hypertension. Technological advancements in medicine have also helped extend lifespans and improve quality of life for people worldwide.

- **Productivity**

Technology has also vastly increased productivity. The ability of computers to solve complex mathematical equations allows them to speed up any task requiring measurement or other calculations. Computer modeling of physical effects can save time and money in any manufacturing situation, giving engineers the ability to simulate structures, vehicles or materials to provide primary data on performance before prototyping. Even in the office environment, the ability of networked computers to share and manipulate data can speed a wide variety of tasks, allowing employees to work together efficiently for maximum productivity.

There are many uses of technologies in our daily life. Now a day's people use technologies to do different kinds of tasks. This is very significant for young, older and anyone who is taking benefit for modern technology, and how it will affect on us, and how it can improve our lives or other people lives. However, what is essential about modern technology which makes a real diversity in our lives. Its verification that people are using internet to receive information, especially about current or main events, and Majority of people surveyed, who is happy with their skills by using internet technology.
Modern Technologies play a very important role in our lives. For instance, modern use of technology increases efficiency. People have always used modern technological devices for development of a culture to solve problems; in earlier days of life ancient people have taken benefits of writing to communicate an idea with someone. They used to draw pictures to find solutions to public issues on their cave walls. If we look back to early human history when the ancient people recorded and spoken words were used to convey information from one generation to the next generation. Now a day's Modern technology helping us for doing work with less effort, even though it's helping us to passes knowledge to groups and individuals. So, people can develop self regulatory skills to increased more efficiency and accomplishing simple to complex everyday tasks. Now, people can also make an appointment with their doctor by using online modern technology. The uses of modern online technologies like uses of internet, television, radio, laptop, cell phones are tremendously revolutionized the world. We can also pay our bills without going somewhere

Advantages and disadvantages of using new technology for businesses

New technology has a range of advantages and disadvantages for businesses and business stakeholders. It is important for businesses to assess the risk and make informed decisions about whether to use the latest technology.

Advantages of new technology include:

- easier, faster and more effective communication
- better, more efficient manufacturing techniques
- less wastage
- more efficient stock management and ordering systems
- the ability to develop new, innovative approaches
- more effective marketing and promotion
- new sales avenues

Disadvantages of new technology include:

- increased dependency on technology
- often large costs involved with using the latest technology (especially for small businesses)
- increased risk of job cuts
- closure of high street stores in favour of online business
- security risk in relation to data and fraud
- required regular updates
- can go down or have faults, which can stop all business operations instantly.

https://www.techquintal.com/modern-technology/

What is Modern Technology?

Modern technology is the advancement of old technology with new additions and modifications. For example, it is impossible for the people in this decade to live with a wired telephone placed on the table. So, the mobile phone which can be taken with us anywhere is the perfect example of technological advancement or simply, modern technology. Any machine or device we see around us is the product of modern technology. It made things way easier than we could ever imagine.

Modern technology has impacted every area of our lives and become a crucial factor in no time. We even reached a point where everything we use in our daily life is directly or indirectly related to the technological advancement of some form. So, modern technology cannot be avoided even if we intentionally wanted to.

It is all about doing things in a much quicker, efficient way by improving the workflow. Either it could be helping humans or doing the task alone, machines are always better in terms of accuracy and efficiency. So, we are taking advantage of them to make our jobs easier. The realization that we can do a lot more stuff within minimal time, with less effort paved the path to modern technology as we see today.

Examples of Modern Technology

The latest modern technology examples include the 5G network which provides blazing fast internet to the users, self-driving cars, and reusable satellite launchers. But it is not limited to just these things. The technological progress we have made and the number of tools we have invented is beyond imagination. Here is a list of modern technology inventions that turned our lives easier and enhanced the easiness of performing operations in every industry. These are the **examples of modern technology**.

1. VR Headsets

In this modernized tech world, the place of VR headsets is no more hidden. The virtual world is directly taken in front of your eyes by just wearing a box-shaped headset. As you can see in the above image, the VR headsets are very convenient and easily available in the stores. Here is a funny video prank on a man who wore the VR headset, and a rollercoaster is taken in front of his eyes. Watching it will give you an exact idea about the power of VR headsets in tricking your eyes and mind.

2. Smart watches

Smart-watches are just a variation of regular watches but come with a bunch of features as your smart-phone. You can make calls, text your friends, browse the internet, and even take photos sometimes. In the **contemporary tech world**, smart-watches do stand a chance in increasing the productivity of users. Instead of taking out the smart phone each time to answer a call, or text a friend, it is way much easier to use your own watch. The Apple Watch Series 2 and Samsung Gear S3 are some best examples of smart-watches.

3. Robots

Robotics improved a lot in the last few years. Numerous **new inventions** arrived on the market in various forms and for various uses. The robots also widened the possibility of modern technology in education. They are being used to assist the teachers, help students, clean blackboards, clean the classrooms, etc. in hi-tech countries like Japan. They also invented robots that answer to humans utilizing **artificial intelligence**. Look at the following video to get a better idea about how far we reached in robotic.

4. Bitcoin

While modern technology gadgets are discussed above, it is time for something different. Bitcoin might be the best choice when it comes to the digital tech world. **Bitcoin is a virtual money** created by a person who is still not recognized completely. To understand the importance of Bitcoin you have to hear the present-day rate of it. 1 Bitcoin = 896.15 US Dollars as per 01/25/2017. Once it reached even around 1100+. But one of the major disadvantages of Bitcoin is the randomly changing value. The below-given video will provide more information.

5. Self-Driving Cars

Self-driving cars are another great example of modern technology inventions. Companies like Tesla have been building self-driving cars for a while now and keeps improving them on every new model. You don’t even have to put your hands on the wheel unless of an emergency. The cars will take you to any place marked on the map safely without you even worrying about the traffic. The advanced sensors placed on all sides can not only detect when the car is approaching danger but can also predict the possible issues during the drive. More companies are adapting to self-driving cars and ditching their old technologies.

6. Cloud Technology

Cloud technology is growing rapidly. From hosting websites online to developing artificial intelligence, cloud technology is used everywhere. From the traditional hosting technologies, cloud hosting/cloud computing is significantly different due to its extreme flexibility and expandable features. Cloud technology is being utilized in several areas of the industry such as factories, businesses, research organizations, government organizations, etc. to make their jobs easier. Cloud is the future.

Advantages of Modern Technology

There are a lot of advantages and disadvantages of modern technology. However, it seems like, the advantages can hide the impact of the problems caused due to technological development. The advancement in modern automotive technology contributed a significant improvement to humankind and the earth itself just like in every other sector we can imagine.

As we already discussed in the above sections, technology is something which we can't avoid in our life. If you are looking for the merits and demerits of modern technology, here are a few of them which are more oriented to modern technology than the overall tech. These points explain why we can't live without technology.

These are the advantages of modern technology.

1. Modern technology takes Innovation and Creativity to the next level.
2. It makes it much convenient to learn and grab information.
3. Communications feature improved.
4. Increased efficiency of people to complete particular tasks.
5. Huge impact in the education industry.
6. Connected people together through social networking.
7. Transportation facilities boosted productivity.
8. The lifestyle became easier.

9. Improved diagnosis and curing equipment enhanced the health industry.
10. Entertainment gadgets improved to the extreme.
11. Improved the problem-solving ease.
12. Helped small businesses to grow faster.
13. Numerous equipment got lesser prices with new technologies.

Disadvantages of Modern Technology

Modern technology got some disadvantages too. Like anything in the world which has a good side and a bad, modern technology also has the same. As you may have already guessed, the demerits aren't any less. These are the disadvantages of modern technology.

1. Modern technology creates job insecurity due to the excessive use of robots and machines.
2. It increased competition in every field due to the usage of automation.
3. Affected social life by keeping people attached to tech gadgets.
4. Creation of harmful weapons and machinery.
5. Increased pollution of air, water, and soil.
6. Cybercrimes increased, and almost anything is destructible by brilliant hackers.
7. Modern technology can create health complications like obesity due to the addiction to devices such as smartphones or tablets.
8. It can be a huge waste of time due to addiction to modern gadgets.
9. It can cause distractions from studies and other natural activities.
10. Modern technology can affect the creativity of people because of the easiness to do things using technology.

Applications of Modern Technology

As we know, technology isn't dedicated to a single sector but is spread across multiple areas including Education, Business, Medicine, etc. Now, let's see how modern technology has improved these sectors from what it used to be in the past. These are the **applications of modern technology** in different sectors.

1. Modern Technology in Education

We already spoke about how modern technology helps education in-depth. Let's mention a few obvious benefits to give you a short reminder. You can receive *instant notifications, weekly schedules, lectures, practice tests, educational games, or presentations* to your smartphone, computer, or even a smartwatch. Then, you can use text, audio, and video to communicate back and forth with other students and professors. Furthermore, modern technology allows you to never step foot in a brick-and-mortar school again. You can earn your credits online and receive your diploma via mail.

The environment benefits simultaneously. Professors and authors no longer need to print thousands of books to spread knowledge. A small electronic file, a low-spec computer, and a low-cost projector in person, or screen sharing through the Internet can help educate a large group of students for cheap. The reduced cost of operation also benefits the pupils. Many schools and universities use the money to provide scholarships and purchase equipment for impoverished students. That way, everyone has an equal opportunity to learn. Purdue University has a great article on the effect of technology on education.

2. Modern Technology in Business

Remember long commute, shared offices filled with cramped cubicles, and long car/airplane rides just to close a business deal, do market research, or handle bureaucracy? Modern technology has provided an alternative for all of those and more. You can do many of the **jobs from home**, removing transportation costs and huge time waste. If the job requires you to go outside, you can receive instructions on your smartphone or smartwatch. You can use the same device to do research, communicate with the business partners, and in some cases, handle the entire business.

The ability to communicate, delegate, analyze, and determine things remotely improves efficiency, productivity, and chances of success. Plus, every such device has a built-in GPS and an Internet connection. This helps human resources, marketing, production, and management departments evaluate progress, get real-time feedback, and ensure everyone is doing their part correctly.

In turn, this **positively affects workers**‘ attitude, interaction, satisfaction, and improves their safety. Speaking about safety and security, you can’t help but think of the difference some of the technological advancements have made. Think of encryption, Information Technology, storage space, remote access, high-quality cameras, motion sensors, fingerprint sensors, VPNs, and other safeguards.

Perhaps the biggest advantage modern technology has brought to business is **globalization**. A tiny company can compete with an industry giant and reach a market across the globe. This affects multiple things positively. It improves competition and allows companies to find remote sources for raw materials, workforce, knowledge, improvements, and opportunities. Best of all, the company doesn’t need to have a physical store. Thanks to modern technology, and e-commerce popularity, more types of businesses are making a shift to cut operating costs.

3. Modern Technology in Medicine

et’s begin with disease research and the accelerated spreading of information in the event of a pandemic. It has shown us that there’s an alternative to doing real-life tests and risking money, health, and wasting time. Instead, various things can be repeatedly simulated, observed, and drawn conclusions from thanks to powerful GPUs and CPUs. And, thanks to Virtual Reality (VR) and Augmented Reality (AR), doctors can run tests and practice in a computer-generated environment without a huge financial toll. They also won’t have to risk human life or be afraid of making a mistake.

Generating a **realistic 3D model** also saves time, allows for indefinite tests and tweaks, and can reduce manufacturing costs. Think of artificial limbs, joints, and even skin. Some are so advanced that they respond to brain signals and might start to imitate nerve ends. Modern technology also helps people with a loss of one or more senses. Instead of being shunned, powerless, and relying on others, they can now communicate and educate themselves. Most importantly, they can live a long, comfortable, and productive life.

The patient files used to get lost, altered at the doctor's discretion, and even falsified. Now, because there are multiple backups, they are both impervious to damage, and the history of edits is clearly visible. On the other hand, anyone with the right credentials can access them at a moment's notice, even halfway across the globe.

Doctors can also use a vast, constantly updated database of knowledge. This allows them to learn about the newest developments and choose the right treatment. If needed, they can consult with other professionals, regardless of distance or time zone. *Modern machines can also monitor vital functions*, send real-time feedback, and even suggest treatment, administer medicine, or automate processes such as 24/7 life support or hospice care. Pharmaceutical technology is also a great advancement in the field of medical technology.

4. Modern Technology in Agriculture

Agriculture has been there since the development of humankind as a species has started. From the time people started realizing that they can grow food in the soil, they have been trying things. Different crops, seeds, cultivation methods, and finally, incorporating technology to make everything easier. Instead of manually plowing acres of land with the limited power of a human, a tractor can make it easier like never before. The results from hundreds of human hours can be achieved within a few hours on a tractor.

It is just an example of how modern technology has enhanced the production of food and the effectiveness of agriculture. From the cultivation of the seeds to storing them under appropriate conditions, everything is done using precise measurements and knowledge achieved from in-depth researches conducted with the help of technology. Controlling pests and making the food items grow faster saving a ton of time made everything much more effective.

Scientists are even producing hybrids of plans adapting the good factors of different species, resulting in a much better form. Not talking about the budding and crafting we all learned in the schools. But there is much more going on. None of these would be possible if we didn't have the help of highly capable computers and tons of technological resources around us. So, modern technology definitely has a huge impact on agriculture.

New Technology

Listed Below Are the Top New Technology Trends, 2022.

1. Computing Power
2. Smarter Devices
3. Quantum Computing
4. Datafication
5. Artificial Intelligence and Machine Learning
6. Extended Reality
7. Digital Trust
8. 3D Printing
9. Genomics
10. New Energy Solutions
11. Robotic Process Automation (RPA)
12. Edge Computing
13. Quantum Computing
14. Virtual Reality and Augmented Reality
15. Blockchain
16. Internet of Things (IoT)
17. 5G
18. Cyber Security

Technology is the way we apply scientific knowledge for practical purposes. It includes machines (like computers) but also techniques and processes (like the way we produce computer chips). It might seem like all technology is only electronic, but that's just most modern technology. In fact, a hammer and the wheel are two examples of early human technology.

Technology in Everyday Life

Let's consider some examples of how technology is integral to our daily lives. When you get up in the morning, you probably get out of a bed. The synthetic materials of the mattress upon which you were sleeping, and springs underneath, are both examples of technology.

If it's still early, you might turn on the light first. Both the light bulbs and the electrical systems that power them are also technology. Later, when you brush your teeth, the system that brings you water to the sink, the bathroom fan, the toothbrush - and the toilet, for that matter - are technology.

If you're like many millions of people, you probably turn on the computer pretty quickly after waking. A computer is one of the most advanced pieces of technology we've ever come up with as humans, and the processes of making the computer's parts are all also technology.

It would be impossible to list every single example of technology in our daily lives. Whether it's practical (like washing machines, tumble dryers, refrigerators, cars, flooring materials, windows, or door handles) or for leisure (like televisions, Blu-ray players, games consoles, reclining chairs, or toys), all these things are examples of technology.

Technology is the application of scientific knowledge to the practical aims of human life or, as it is sometimes phrased, to the change and manipulation of the human environment.

History of E-Commerce

- The history of Ecommerce seems rather short but its journey started over 40 years ago in hushed science labs.
- In the 1960s , very early on in the history of Ecommerce, its purpose was to exchange long distance electronic data. In these early days of Ecommerce, users consisted of only very large companies, such as banks and military departments, who used it for command control communication purposes. This was called EDI, and was used for electronic data interchange.
- Originally, electronic commerce was identified as the facilitation of commercial transactions electronically, using technology such as Electronic Data Interchange (EDI) and Electronic Funds Transfer (EFT). These were both introduced in the late 1970s, allowing businesses to send commercial documents like purchase orders or invoices electronically.
- The growth and acceptance of credit cards, automated teller machines (ATM) and telephone banking in the 1980s were also forms of electronic commerce
- In 1982 Transmission Control Protocol and Internet Protocol known as TCP & IP was developed. This was the first system to send information in small packets along different routes using packet switching technology, like today's Internet! As opposed to sending the information streaming down one route
- Beginning in the 1990s, electronic commerce would include enterprise resource planning systems (ERP), data mining and data warehousing
- In 1995, with the introduction of online payment methods, two companies that we all know of today took their first steps into the world of Ecommerce. Today Amazon and ebay are both amongst the most successful companies on the Internet.

Year	Event
1984	EDI, or electronic data interchange, was standardized through ASC X12. This guaranteed that companies would be able to complete transactions with one another reliably.
1992	Compuserve offers online retail products to its customers. This gives people the first chance to buy things off their computer.
1994	Netscape arrived. Providing users a simple browser to surf the Internet and a safe online transaction technology called Secure Sockets Layer.
1995	Two of the biggest names in e-commerce are launched: Amazon.com and eBay.com.
1998	 DSL, or Digital Subscriber Line, provides fast, always-on Internet service to subscribers across California. This prompts people to spend more time, and money, online.
1999	Retail spending over the Internet reaches $20 billion, according to Business.com.
2000	The U.S government extended the moratorium on Internet taxes until at least 2005.

Assignments

1.Define the term "contemporary technology" and "old technology" .Write any five examples.

2.Why Contemporary technology is needed ? Explain

or

Explain the importance of contemporary technology.

3.What are the advantages and disadvantages of using new technology or contemporary technology or modern technology ?

4.Explain the applications of contemporary technology .

5.List out any 10 presently used contemporary technologies . Explain any five contemporary technologies.

Unit-2 : E-Commerce

E-Commerce Definition

- **E-Commerce or Electronic Commerce means buying and selling of goods, products, or services over the internet.** E-commerce is also known as electronic commerce or internet commerce. These services provided online over the internet network.
- Transaction of money, funds, and data are also considered as E-commerce.
- These business transactions can be done in four ways: Business to Business (B2B), Business to Customer (B2C), Customer to Customer (C2C), Customer to Business (C2B).
- The standard definition of E-commerce is a commercial transaction which is happened over the internet.

Online stores like Amazon, Flipkart, Shopify, Myntra, Ebay, Quikr, Olx are examples of E-commerce websites. By 2020, global retail e-commerce can reach up to $27 Trillion.

Introduction to Commerce

- Commerce is basically an economic activity involving trading or the buying and selling of goods.
- For e.g. a customer enters a book shop, examines the books, select a book and pays for it. To fulfill the customer requirement, the book shop needs to carry out other commercial transactions and business functions such as managing the supply chain, providing logistic support, handling payments etc.
- As we enter the electronic age, an obvious question is whether these commercial transactions and business functions can be carried out electronically.In general, this means that no paperwork is involved, nor is any physical contact necessary. This often referred to as electronic commerce (e-commerce).
- The earliest example of e-commerce is electronic funds transfer. This allows financial institutions to transfer funds between one another in a secure and efficient manner.
- Later, electronic data interchange (EDI) was introduced to facilitate inter-business transactions.

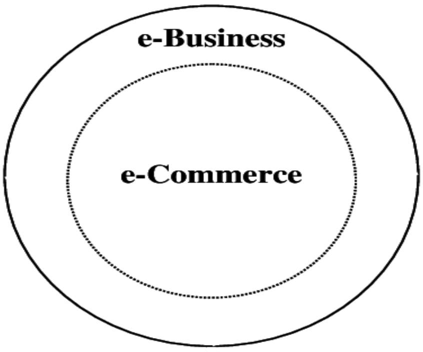
E-Commerce definition

- “E-Commerce or Electronic Commerce, a subset of E-Business, is the purchasing, selling and exchanging of goods and services over computer networks (such as Internet) through which transactions are performed”.
- “E-Commerce can be defined as a modern business methodology that addresses the needs of organizations, merchants and consumers to cut costs while improving the quality of goods and services and increasing the speed of service delivery by using Internet”.

- ❖ E-Commerce takes place between companies, between companies and their customers, or between companies and public administration.
- ❖ FEW EXAMPLES OF E-Commerce are:Amazon.com, an online bookstore started in 1995 grew its revenue to more than 600$ million in 1998.Microsoft Expedia, an integrated online travel transaction site helps to choose a flight, buy an airline ticket, book a hotel, rent a car etc. in only a few minutes.

E-Business enables organizations to accomplish the following goals:

- Reach new markets.
- Create new products or services.
- Build customer loyalty
- Make the best use of existing and emerging technologies.
- Achieve market leadership and competitive advantage.
- Enrich human capital.



e-Commerce is a subset of e-Business

Differences between Traditional Commerce and E-commerce

1. Traditional Commerce :

Traditional commerce refers to the commercial transactions or exchange of information, buying or selling product/services from person to person without use of internet which is a older method of business style and comes under traditional business. Now a days people are not preferring this as it is time taking and needs physical way of doing business.

Example includes physical market/bazaar.

2. E-commerce :

E-commerce refers to the commercial transactions or exchange of information, buying or selling product/services electronically with the help of internet which is a newer concept of business style and comes under e-business. Now a days people are preferring this as it is less time taking and does not need physical way of doing business everything can be done with laptop or smartphone and internet.

Example includes online shopping sites.

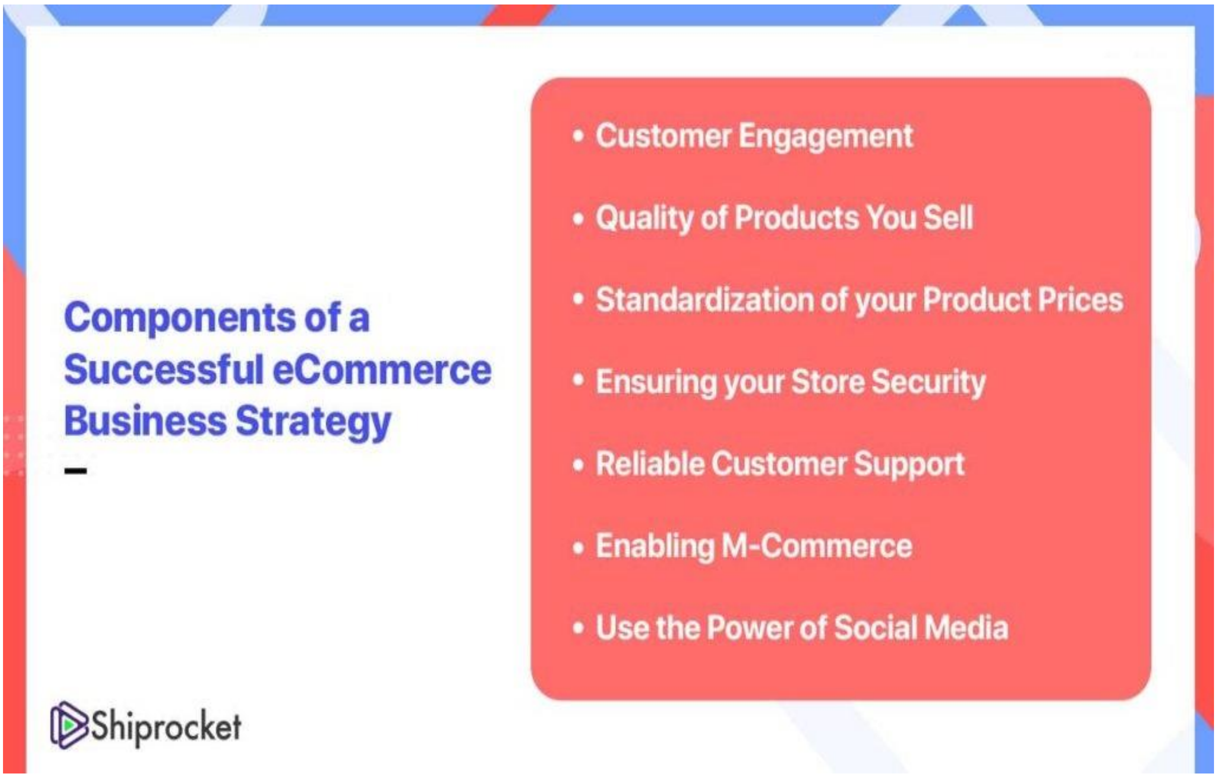
S.No.	TRADITIONAL COMMERCE	E-COMMERCE
01.	Traditional commerce refers to the commercial transactions or exchange of information, buying or selling product/services from person to person without use of internet.	E-commerce refers to the commercial transactions or exchange of information, buying or selling product/services electronically with the help of internet.
02.	In traditional commerce it is difficult to establish and maintain standard practices.	In e-commerce it is easy to establish and maintain standard practices.
03.	In traditional commerce direct interaction through seller and buyer is present.	In e-commerce indirect interaction through seller and buyer occurs using electronic medium and internet.
04.	Traditional commerce is carried out by face to face, telephone lines or mail systems.	E-commerce is carried out by internet or other network communication technology.
05.	In traditional commerce processing of transaction is manual.	In e-commerce processing of transaction is automatic.
06.	In traditional commerce delivery of goods is instant.	In e-commerce delivery of goods takes time.
07.	Its accessibility is for limited time in a day.	Its accessibility is 24×7×365 means round the clock.
08.	Traditional commerce is done where digital	E-commerce is used to save valuable time

	network is not reachable.	and money.
09.	Traditional commerce is a older method of business style which comes under traditional business.	E-commerce is a newer concept of business style which comes under e-business.
10.	Its resource focuses on supply side.	Its resource focuses on demand side.
11.	In traditional commerce customers can inspect products physically before purchase.	In e-commerce customers can not inspect products physically before purchase.
12.	Its business scope of business is a limited physical area.	Its business scope is worldwide as it is done through digital medium.
13.	For customer support, information exchange there is no such uniform platform.	For customer support, information exchange there is exists uniform platform.

Components of E-Commerce

Every business person is looking to have an online store to sell their range of products and services. Creating an effective business strategy is one of the major keys as it defines what your business offers. Having a clear purpose and direction will help you understand the things you need and your consumers.

Having an efficient e-Commerce business strategy is vital in cutting down on possible unwanted expenses and losses. This also assists you in competing with other bigger competitors in the online retail industry.



Customer Engagement

To convert potential consumers, you need to make a good first impression. Your website represents your online presence, and you need to be creative as well.

Getting the best design for your website can last your impression, and easy navigation to your website can give your consumers an excellent shopping experience. You can go for a less stylish design or bold themes with colors to represent your website.

The pages of your website are also important. Your "About Us" page conveys details about your business niche, your location, and what you serve. Adding good quality images and a unique description of your products is also a must. Adding an FAQ page that can answer the basic questions of your customers will add up to your customer satisfaction experience.

Quality Of Your Products

The quality of your products can help you gain trusted and loyal consumers. This also reduces your time, cost, and risk of getting return requests for defective products.

This will create a negative impression that products you sell online are not of good quality. You create a good impression and brand reputation by ensuring your online consumers that your products are genuine and of good quality. Getting ISO accreditation is another way of gaining your consumers' trust.

Standardization Of Your Product Prices

Online consumers always look for and compare the product prices that you offer. Product pricing is considered to be a marketing tool and has a direct effect on your conversion rates. This is the reason why you must evaluate your product prices.

You have to keep in mind that when an online consumer visits a website, the first thing they look for is the product's price A proven way to standardize your product pricing is by having a cost-based model that works in three steps deciding your cost price, wholesale price, and your retail price.

By standardizing your product pricing strategy, you will always succeed in the type of online retail business you have.

Ensuring Your Store Security

Your online store must be secured so that your consumers will have the confidence to purchase your products. Your online retail shopping cart should have excellent security features that will prevent your consumers' personal information from falling into the wrong hands. This can directly improve your business' reputation.

There are different ways to secure your eCommerce website. One is by using an SSL (Secure Socket Layer) certificate that encrypts data on your website to secure it from online threats. Another is implementing advanced verification methods. This will earn your consumer's trust because they know you are taking good security measures.

Reliable and Trusted Customer Support

Online consumers trust the reliability part. If your customer support attends to their questions, queries, and problems relating to product purchase, payment, returns, and delivery adds reliability and trust to your brand.

Your customer service should be available 24/7 and must provide your consumers with an excellent approach. Additionally, you can use a chatbot for connecting with your audience on a personal level.

Having good customer service will help you gain and retain consumers. This will also help you build your brand identity.

Enabling M-Commerce

Your online eCommerce business should be able to adapt to technological advancement to stay competitive in the market. M-Commerce or Mobile Commerce is an innovative trend in online retailers nowadays.

Most consumers do their product research, purchase, and payment with the use of their smartphones. Having a mobile app for your online retail store is the main component of an effective eCommerce business. It is also important that you should monitor and upgrade your app regularly.

Utilize The Power Of Social Media

There are approximately 4.4 billion internet users worldwide, out of which 3.44 billion are active on social media. Most companies sell products or services on Instagram Shopping as it generates more attention and engagement than Facebook.

Therefore, promoting your eCommerce business on social media will surely boost your store's presence, engagement, and conversion rate.

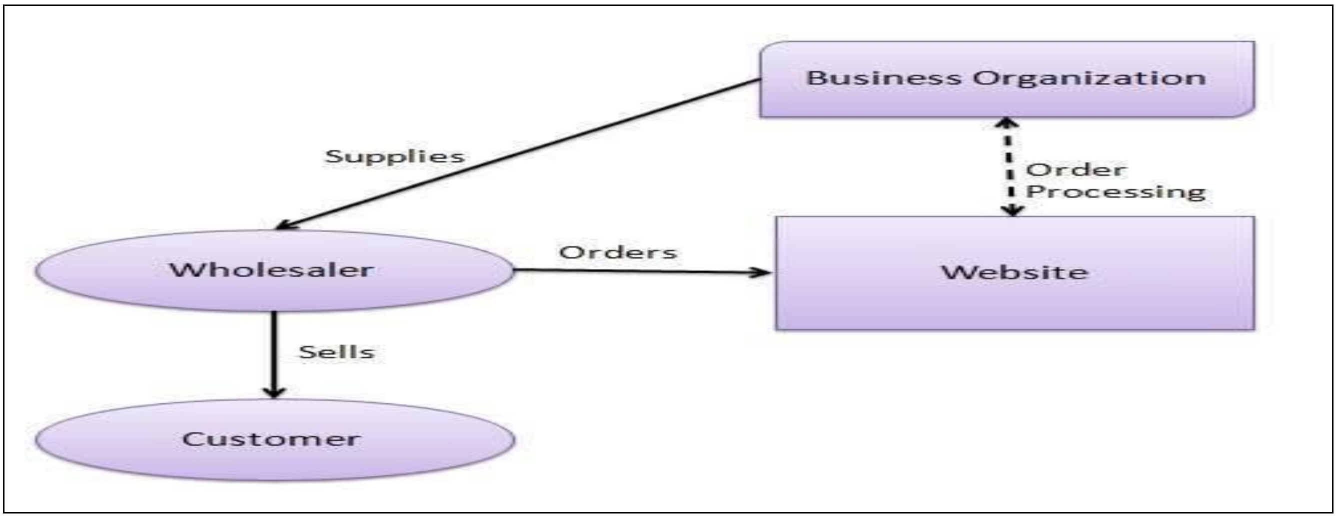
Types of E-Commerce

- **E-commerce business models can generally be categorized into the following categories.**

1. Business - to - Business (B2B)
2. Business - to - Consumer (B2C)
3. Consumer - to - Consumer (C2C)
4. Consumer - to - Business (C2B)
5. Business - to - Government (B2G)
6. Government - to - Business (G2B)
7. Government - to - Citizen (G2C)

1.Business -to- Business

A website following the B2B business model sells its products to an intermediate buyer who then sells the product to the final customer. As an example, a wholesaler places an order from a company's website and after receiving the consignment, sells the end product to the final customer who comes to buy the product at one of its retail outlets.



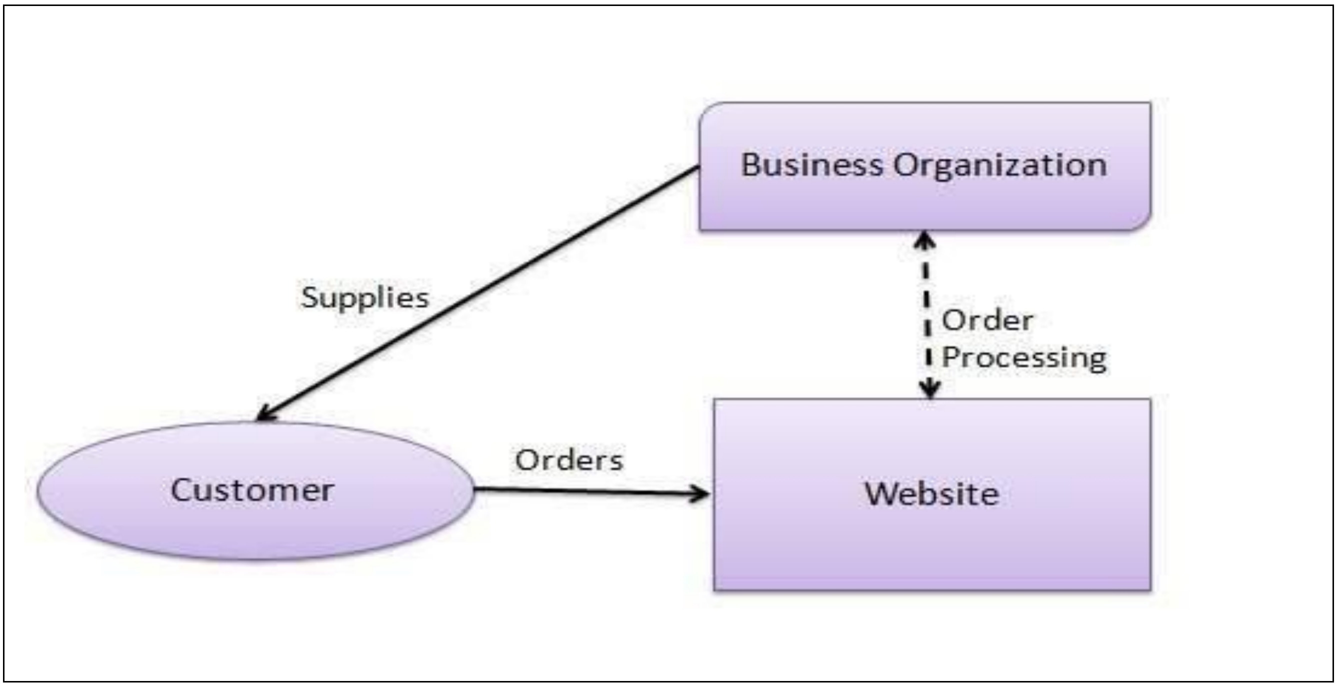
- ❖ B2B is one of the most common types of e-commerce. This is when a transaction of goods or services occurs between two businesses.
- ❖ B2B e-commerce is simply defines as e-commerce between Companies .This is the type of e-Commerce that deals with relationship between and among businesses. examples- Indiamart ,Alibaba ,Amazon Business.

Business -to-Business (B2B)

- ❖ **Business to Business or B2B refers to e-commerce activities between businesses. An E-Commerce company can be dealing with suppliers or distributers or agents.**
- ❖ These transactions are usually carried out through Electronic Data Interchange (EDI). EDI is an automated format of exchanging information between businesses over private networks.
- ❖ For e.g. manufacturers and wholesalers are B2B Companies.
- ❖ By processing payments electronically, companies are able to lower the number of clerical errors and increase the speed of processing invoices, which result in lowered transaction fees.
- ❖ In general, B2Bs require higher security needs than B2Cs.
- ❖ With the help of B2B E-commerce, companies are able to improve the efficiency of several common business functions, including supplier management, inventory management and payment management.

2.Business - to - Consumer

- ❖ A website following the B2C business model sells its products directly to a customer. A customer can view the products shown on the website. The customer can choose a product and order the same. The website will then send a notification to the business organization via email and the organization will dispatch the product/goods to the customer.



Business-to-Consumer (B2C)

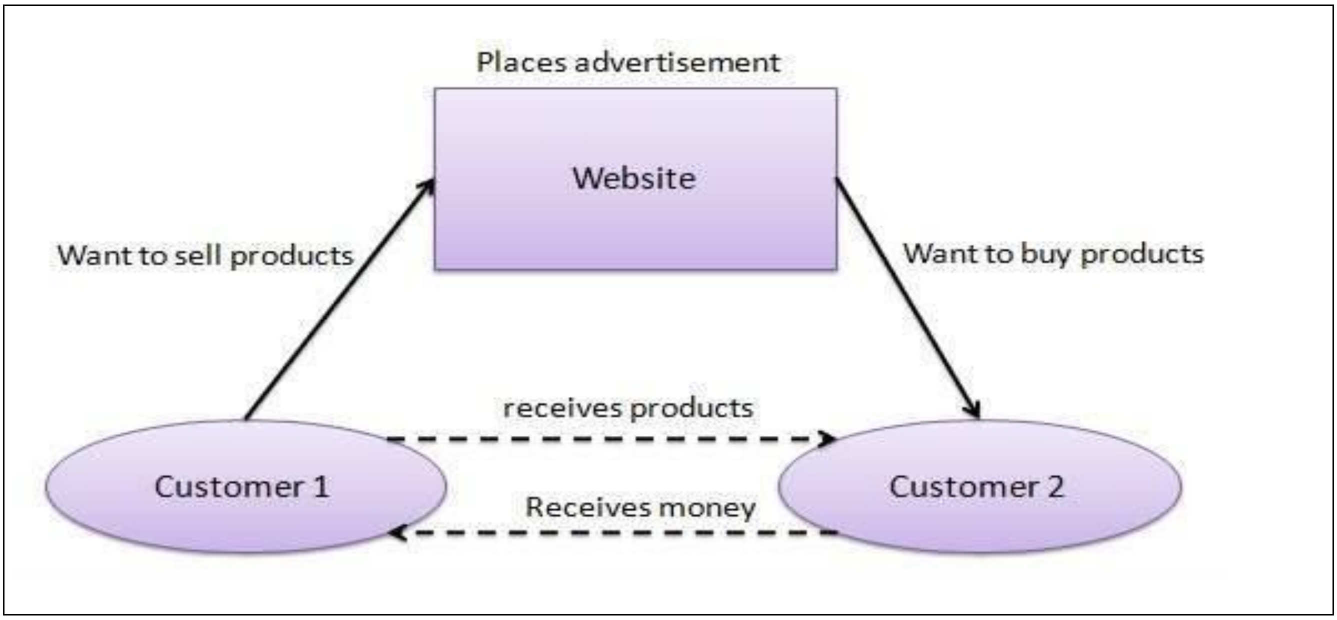
- ❖ Business to Consumer. Here the company will sell their goods and/or services directly to the consumer. The consumer can browse their websites and look at products, pictures, read reviews. Then they place their order and the company ships the goods directly to them. Popular examples are Amazon, Flipkart, Jabong , paying and using Netflix at home. *McDonalds (most big brands).*

Business To Customer (B2C):-

- ❖ Business to Customer or B2C refers to E-Commerce activities that are focused on consumers rather than on businesses.For instance, a book retailer would be a B2C company such as Amazon.com. Other examples could also be purchasing services from an insurance company, conducting on-line banking and employing travel services.

3.Consumer - to - Consumer

- ❖ A website following the C2C business model helps consumers to sell their assets like residential property, cars, motorcycles, etc., or rent a room by publishing their information on the website. Website may or may not charge the consumer for its services. Another consumer may opt to buy the product of the first customer by viewing the post/advertisement on the website.



Consumer-to Consumer (C2C)

- Consumer -to -consumer e-commerce or C2C is simply commerce betwwen private individuals or consumers.Consumer to consumer, where the consumers are in direct contact with each other. No company is involved. It helps people sell their personal goods and assets directly to an interested party. example eBay,Olx,Quikr

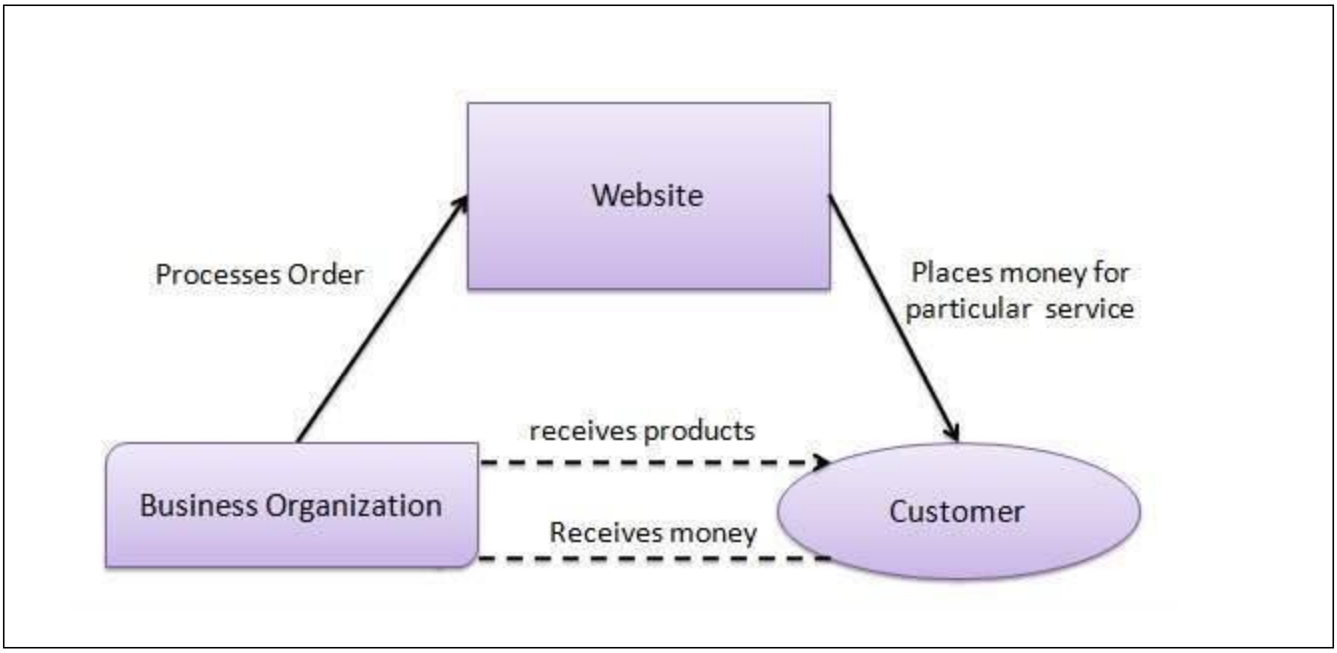
Customer To Customer (C2C):-

- Customer to Customer or C2C refers to E-commerce activities, which use an auction style model. This model consists of a person-to-person transaction that completely excludes businesses from the equation.

Customers are also a part of the business and C2C enables customers to directly deal with each other. An example of this is peer auction giant ebay.

4.Consumer - to - Business

- In this model, a consumer approaches a website showing multiple business organizations for a particular service. The consumer places an estimate of amount he/she wants to spend for a particular service. For example, the comparison of interest rates of personal loan/car loan provided by various banks via websites. A business organization who fulfills the consumer's requirement within the specified budget, approaches the customer and provides its services.



Consumer -to-Business (C2B)

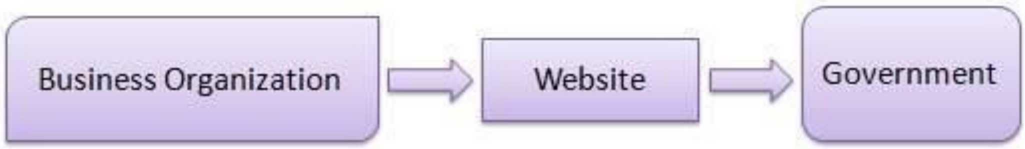
- The consumer to business model isn't all that traditional. The idea here is that businesses are buying from solo consumers, although more often than not those consumers are operating a business of their own.This is the reverse of B2C, it is a consumer to business. So the consumer provides a good or some service to the Company. Say for example an IT freelancer who demos and sells his software to a company. This would be a C2B transaction

Customer To Business (C2B)

- Customer to Business or C2B refers to E-Commerce activities which use reverse pricing models where the customer determines the prices of the product or services.In this case, the focus shifts from selling to buying. There is an increased emphasis on customer empowerment.In this type of E-Commerce, consumers get a choice of a wide variety of commodities and services, along with the opportunity to specify the range of prices they can afford or are willing to pay for a particular item, service or commodity.

5.Business-to -Government

- B2G model is a variant of B2B model. Such websites are used by governments to trade and exchange information with various business organizations. Such websites are accredited by the government and provide a medium to businesses to submit application forms to the government.



Business -to -Government (B2G)

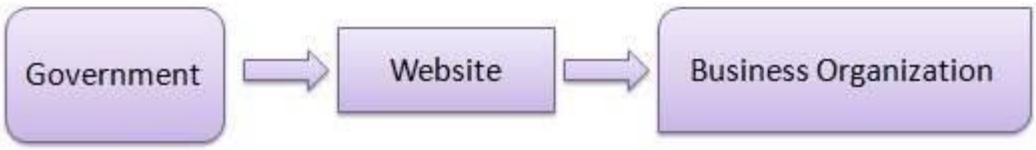
B2G e-commerce is commerce between Companies and public sector. it refers to the use of the Internet for Public Procurement, licensing procedure, and other government-related Operation.Example: — Business pay taxes, file reports, or sell goods and services to Govt. agencies.

Business To Government (B2G)

- It is a new trend in E-Commerce. This type of E-Commerce is used by the government departments to directly reach to the citizens by setting up the websites.These websites have government policies, rules and regulations related to the respective departments.Any citizen may interact with these websites to know the various details. This helps the people to know the facts without going to the respective departments.This also saves time of the employees as well as the citizens.

6.Government - to - Business

- Governments use B2G model websites to approach business organizations. Such websites support auctions, tenders, and application submission functionalities.



Government- to- Business (G2B)

- G2B e-commerce is a business model were all the information and services are provided by the Government to the Business Organizations. The information is shared through a vast network of different government websites.The business Organization use that information to apply for various permission needed for starting a new business, and other specifications.A

Business Organization can also download the different forms and submit it online or offline to the related office. Example if you visit https://www.maharashtra.gov.in you will be able to read all the important Information about current tenders.

7.Government - to - Citizen

❖ Governments use G2C model websites to approach citizen in general. Such websites support auctions of vehicles, machinery, or any other material. Such website also provides services like registration for birth, marriage or death certificates. The main objective of G2C websites is to reduce the average time for fulfilling citizen's requests for various government services.

Government -to -Consumer (G2C)

❖ The **electronic commerce** activities performed between the government and its citizens or consumers, including paying taxes, registering vehicles, and providing information and services,registration for birth, marriage or death certificates.

8.Government -to -Government (G2G)

❖ Government to government (G2G) is the electronic sharing of data and/or information systems between government agencies, departments or organizations. The goal of G2G is to support e-government initiatives by improving communication, data access and data sharing.Any E-commerce transaction happening between 2 or more governments or within departments of government .

Mobile Commerce (M -Commerce)

❖ **M-commerce (Mobile Commerce) is the buying and selling of goods and services through wireless handheld devices such as smartphones and tablets.**

❖ Experts consider mobile commerce as the next phase of eCommerce, as it allows consumers to buy goods or services online — but from anywhere and at any time.But mobile commerce is so much more than that. In fact, M- Commerce triggered the emergence of brand-new industries and services or helped the existing ones to grow in new directions.

❖ **Examples of such innovations include:**

- tickets and boarding passes,
- mobile banking,
- money transfers, contactless payments, and in-app payments,
- digital content purchases,
- location-based services,
- mobile marketing, including coupons and loyalty cards.

All of this M-Commerce features which today are part of our customer experience .

Most common applications of E-commerce:

Retail and Wholesale

Ecommerce has numerous applications in this sector. E-retailing is basically a B2C, and in some cases, a B2B sale of goods and services through online stores designed using virtual shopping carts and electronic catalogs. A subset of retail ecommerce is m-commerce, or mobile commerce, wherein a consumer purchases goods and services using their mobile device through the mobile optimized site of the retailer. These retailers use the E-payment method: they accept payment through credit or debit cards, online wallets or internet banking, without printing paper invoices or receipts.

Online Marketing

This refers to the gathering of data about consumer behaviors, preferences, needs, buying patterns and so on. It helps marketing activities like fixing price, negotiating, enhancing product features, and building strong customer relationships as this data can be leveraged to provide customers a tailored and enhanced purchase experience.

Finance

Banks and other financial institutions are using e-commerce to a significant extent. Customers can check account balances, transfer money to other accounts held by them or others, pay bills through internet banking, pay insurance premiums, and so on. Individuals can also carry out trading in stocks online, and get information about stocks to trade in from websites that display news, charts, performance reports and analyst ratings of companies.

Manufacturing

Supply chain operations also use ecommerce; usually, a few companies form a group and create an electronic exchange and facilitate purchase and sale of goods, exchange of market information, back office information like inventory control, and so on. This enables the smooth flow of raw materials and finished products among the member companies and also with other businesses.

Online Booking

This is something almost every one of us has done at some time – book hotels, holidays, airline tickets, travel insurance, etc. These bookings and reservations are made possible through an internet booking engine or IBE. It is used the maximum by aviation, tour operations and hotel industry.

Online Publishing

This refers to the digital publication of books, magazines, catalogues, and developing digital libraries.

Digital Advertising

Online advertising uses the internet to deliver promotional material to consumers; it involves a publisher, and an advertiser. The advertiser provides the ads, and the publisher integrates ads into online content. Often there are creative agencies which create the ad and even help in the placement. Different types of ads include banner ads, social media ads, search engine marketing, retargeting, pop-up ads, and so on.

Auctions

Online auctions bring together numerous people from various geographical locations and enable trading of items at negotiated prices, implemented with e-commerce technologies. It enables more people to participate in auctions. Another example of auction is bidding for seats on an airline website – window seats, and those at the front with more leg room generally get sold at a premium, depending on how much a flyer is willing to pay.

E-Commerce is all around us today, and as an entrepreneur, you should also get into this realm if you want to expand your markets, get more customers and increase your profitability.

Applications of E-Commerce

❖ The applications of E-commerce are used in various business areas such as retail and wholesale and manufacturing. The most common E-commerce applications are as follows:

1. Online marketing and purchasing
2. Retail and wholesale
3. Finance
4. Manufacturing
5. Online Auction
6. E-Banking
7. Online publishing
8. Online booking (ticket, seat.etc)

Online marketing and purchasing

Data collection about customer behavior, preferences, needs and buying patterns is possible through Web and E-commerce. This helps marketing activities such as price fixation, negotiation, product feature enhancement and relationship with the customer

Retail and wholesale:

E-commerce has a number of applications in retail and wholesale. E-retailing or on-line retailing is the selling of goods from Business-to-Consumer through electronic storesthat are designed using the electronic catalog and shopping cart model. Cybermall is a single Website that offers different products and services at one Internet location. It attracts the customer and the seller into one virtual space through a Web browser.

Finance:

Financial companies are using E-commerce to a large extent. Customers can check the balances of their savings and loan accounts, transfer money to their other account and pay their bill through on-line banking or E-banking. Another application of E-commerce is on-line stock trading. Many Websites provide access to news, charts, information about company profile and analyst rating on the stocks.

Manufacturing:

E-commerce is also used in the supply chain operations of a company. Some companies form an electronic exchange by providing together buy and sell goods, trade market information and run back office information such as inventory control. This speeds up the flow of raw material and finished goods among the members of the business community. Various issues related to the strategic and competitive issues limit the implementation of the business models. Companies may not trust their competitors and may fear that they will lose trade secrets if they participate in mass electronic exchanges.

Auctions:

Customer-to-Customer E-commerce is direct selling of goods and services among customers. It also includes electronic auctions that involve bidding. Bidding is a special type of auction that allows prospective buyers to bid for an item. For example, airline companies give the customer an opportunity to quote the price for a seat on a specific route on the specified date and time.

E-Banking:

Online banking or E- banking is an electronic payment system that enables customers of a financial institution to conduct financial transactions on a website operated by the institution, Online banking is also referred as internet banking, e-banking, virtual banking and by other terms.

Online publishing:

Electronic publishing (also referred to as e-publishing or digital publishing) includes the digital publication of e-books, digital magazines, and the development of digital libraries and catalogs.

Online booking (ticket, seat.etc)

An **Internet booking engine** (IBE) is an application which helps the travel and tourism industry support reservation through the Internet. It helps consumers to book flights, hotels, holiday packages, insurance and other services online. This is a much needed application for the aviation industry as it has become one of the fastest growing sales channels.

Advantages of E-Commerce

- There are a number of prominent and not-so-obvious advantages for doing business on an online platform. Understanding exactly how e-Commerce works can help individuals leverage them to their and their businesses advantage:

1. **A Larger Market:** E-Commerce allows individuals to reach customers all across the country and all around the world. E-Commerce gives business owners the platform to reach people from the comfort of their homes. The customers can make any purchase anytime and anywhere, and significantly more individuals are getting used to shopping on their mobile devices.
2. **Customer Insights Via Tracking And Analytics:** Whether the businesses are sending the visitors to their eCommerce website via PPC, SEO, ads, or a good old postcard, there is a way of tracking the traffic and the consumers' entire user journey for getting insights into the keywords, marketing message, user experience, pricing strategy, and many more.
3. **Fast Response To The Consumer Trends And The Market Demands:** Especially for the business people who do "drop ship," the logistics, when streamlined, allow these businesses to respond to the market and the trends of eCommerce and demands of the consumers in a lively manner. Business people can also create deals and promotions on the fly for attracting customers and generate more sales.
4. **Lower Cost:** With the advancement of the eCommerce platforms, it has become very affordable and easy to set up and run an eCommerce business with a lower overhead. Business people no longer need to spend a big budget on TV ads or billboards, nor think about personnel and real estate expenses.
5. **More Opportunities For "Selling.":** Business people can only offer a limited amount of information about a product in a physical store. Besides that, eCommerce websites give them the space to include more information like reviews, demo videos, and customer testimonials for helping increased conversion.

6. **Personalised Messaging:** E-Commerce platforms give people in business the opportunity to provide personalised content and product recommendations for registering customers. These targeted communications can help in increasing conversion by showing the most relevant content to the visitor.
7. **Increased Sales Along with Instant Gratification:** For businesses selling digital goods, e-Commerce allows them to deliver products within seconds of placing an order. This satisfies the needs of the consumers for instant gratification and assists increase sales, especially for the low-cost objects that are often known as "impulse buys."
8. **Ability to Scaling Up (Or Down) Quickly Also Unlimited "Shelf Space.":** The growth of any online business is not only limited by the availability of space. Even though logistics might become an issue as one's business grows, it's less of a challenge compared to running any brick-and-mortar store. E-Commerce business owners can choose to scale up or down their operation quickly by taking advantage of the non-ending "shelf space," as a response to the market trends and demands of consumers.

Disadvantages of E-Commerce

Running a business that is e-commerce is not always rainbows and unicorns. There are unique challenges to this business model — learning about them will help business people navigate the choppy waters and avoid common pitfalls.

1. **Lack of Personal Touch:** Some customers appreciate the personal touch they offer when visiting a physical store by interacting with the sales associates. Such personal touch is especially essential for businesses that sell high-end products as customers will want to buy the products and have an excellent experience during the process.
2. **Lack of Tactile Experience:** No matter how good a video is made, customers still can't feel and touch a product. Not to mention, it's never an easy task to deliver a brand experience that could often be including the sense of touch, taste, smell, and sound via the two-dimensionality of any screen.
3. **Product and Price Comparison:** With online shopping, customers can compare several products and find the least price. This forces many businesses to compete on price and reduce their profit margin, reducing the quality of products.
4. **Need for Access to the Internet:** This is obvious, but don't forget that the customers do need access to the Internet before purchasing from any business! As many eCommerce platforms have the features and functionalities which require a high-speed Internet connection for an optimal consumer experience, there's a chance that companies are excluding visitors who have slow internet connections.

5. **Credit Card Fraud:** Credit card frauds are a natural and growing problem for online businesses. It can lead to many chargebacks, which result in the loss of penalties, revenue, and a bad reputation.
6. **IT Security Issues:** More and more organisations and businesses have fallen prey to malicious hackers who have stolen information of the customers from their databases. This could have financial and legal implications, but it also reduces the company's trust.
7. **All the Eggs in One Basket:** E-Commerce businesses rely solely or heavily on their websites. Even just some minutes of downtime or technology glitches could be resulting in a substantial revenue loss and customer dissatisfaction.
8. **Complexity in Regulations, Taxation, and Compliance:** Suppose any online business sells to its consumers in different territories. In that case, they'll have to stick to the regulations in their own countries or states and their consumers' places of residence. This could be creating a lot of complexities in accounting, taxation and compliance.

Advantages	Disadvantages
Order can be placed from anywhere at any time.	There is no guarantee for the quality of products.
Eliminates the operating cost.	Lack of personal touch.
It helps in connecting with people all across the world.	It doesn't give the luxury of trying before buying the item.
Retargets the customers.	Long delivery period.
There is always detailed product information offered.	There is always a concern with security issues.

Comparison Table for Advantages and Disadvantages of E-Commerce

Advantages	Disadvantages
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E-Commerce in Nepal: Current Situation and Future Scope

E-commerce Business in Nepal is growing rapidly like a fire in the forest with certain difficulties and hurdles. Despite of hurdles there is a huge scope of e-commerce Business in future if certain difficulties are to be solved. In this blog, I have discussed about the difficulties and opportunities in E-commerce business.

E-commerce background

In the present world everything is in the internet. You want to buy mobile,laptops,television anything you want just go through internet you will surely get it. World totally started relying in the internet. Even if you don't like the thing you bought just now through internet you can surely get your money back as well. This is how E-commerce business has already covered the world. Any thing can be bought over internet with either online payment or cash on delivery.

Situation of E-commerce in Nepal

As per growth of e-commerce in foreign land its obvious that in Nepal also e-commerce business is growing but still it has not come up-to the point it was meant to be. Many obstacles are there which i will be mentioning in the following paragraph. Many of us are aware or not but most of the buyer from internet in Nepal usually order through social sites like Facebook or Instagram where you can only order but not online payment. Payment can be done only after we receive what we order i.e cash on delivery.

Difficulties of E-commerce in Nepal

Talking about the difficulties of e-commerce in Nepal here i have listed some of the main problems.

Lack of awareness: With lack of basic computer knowledge with major of Nepal's population there is lack of awareness among them that there exists a world where you can get everything online just by using simple e-commerce sites. I think knowledge must be spread out and people must be known about the facilities.

Trust issue: Now there comes a point where a guy who is familiar about e-commerce business steps back due to trust issue. Trust for what? Trust to get genuine products,trust to get product at cheaper rather than costly.

Payment gateway: Now main problem about e-commerce business in Nepal is facing is payment gateway. Let me tell you briefly about payment gateway. Payment gateway is nothing more than paying the price of goods online. In Nepal we don't have system where we could pay online. We are even restricted for doing the international shopping. Until and unless our country's responsible people take a step forward and help dealing with a bank for legal and secure process of payment gateway we Nepalese still have to dream about international shopping. Soon or later it has to come so why not early?

Lack of competitors:When there is no competition at all its obvious that the price will be high.Reason for less competitor is people not taking initiative and lack of risk bearing capacity. It is believed that in Nepal online shopping is expensive and people rather prefer to go in shop and buy rather in internet.

Scope or Future of e-commerce in Nepal

The future of e-commerce business in Nepal is very much brighter. When after finally payment gateway's problem is solved then there start's the rise of scope of e-commerce in Nepal. Since people already have started taking innovative idea of creating a online wallet system through which we are enabled to do many online shopping and online marketing at least locally within the country. There will be many e-commerce sites through which people will be benefited with many aspects. Use of technology will grow rapidly where even we Nepalese can be a creator in technology field as well which is totally in the minimal volume in the present context. I hope technology will grow so rapidly that even groceries will be possible to do online. It's not that wallet system hasn't brought change in e-commerce in Nepal. Obviously people has done good job after creating wallet but still wallet is not enough for international shopping. Until payment gateway's problem will be solved we Nepalese still have to rely on what we have i.e wallet like eSewa, Goji, Khalti, etc.

New e-commerce law in Nepal: Here are 9 key provisions

With globalisation, the world has seen rapid development in e-commerce. Nepal cannot stay isolated from this phenomenon. Consequently, the Nepali market has been witnessing an upswing in e-commerce businesses.

Meanwhile, information technology is booming majorly because of internet penetration in the Nepali market which has reached 77.91 per cent of the total population as of mid-October 2020.

The upswing was further expanded by the spread of the coronavirus which forced people to lock themselves inside the house. In this commercial world, e-commerce is the future market. But, it is full of regulatory complexities regarding the issues such as data privacy, delivery time, consumer protection, mode of digital payment among a few.

The e-commerce market is just unfolding in Nepal. However, it does not have any clear legal provisions and guidelines which has caused inconvenience to the entrepreneurs in starting and operating the business. On the other hand, consumers are not in a position to trust e-commerce businesses as there are no such laws and state agencies that are fully designated for regulating e-commerce.

Recently, the Ministry of Industry, Commerce and Supplies drafted an e-commerce bill with an aim to create, regulate and facilitate online trade in Nepal. The bill is being reviewed by the concerned ministry which shall be passed from parliament and shall be implemented upon approval from the president. It will be the first dedicated e-commerce law in Nepal.

The bill aims at addressing most of the problems the costumers were facing in the sector currently. In a nutshell, here are nine key provisions proposed in the bill:

1. Order to be deemed as contract

Section 7, sub-section 1: If the seller accepts the order placed by the customer, it shall be deemed as contract and the duties as of contract shall be created.

2. Right to cancellation

Section 7, sub-section 3: The customer can cancel the order until the product is dispatched or the service is provided.

3. No charge in case of cancellation

Section 7, sub-section 4: An order cancelled as per section 7, sub-section 3 shall not be charged and if the amount for the product or service is pre-paid, all the amount shall be refunded to the customer.

4. Diverse payment methods

Section 8, sub-section 2: The customer can pay by using any mode of payment; it may be cash, debit/credit card, internet/mobile banking or through cheque.

5. Fixed place and time for delivery

Section 9: The seller shall deliver the goods in a particular place and time which is mentioned while placing an order.

6. Prohibition on alteration

Section 11(a): The goods and services mentioned on the website shall be delivered to the customers.

7. Provision of replacement and refund

Section 11(b): If the goods delivered is found to be defective, the seller shall replace the goods or shall refund the amount.

8. Non-compliant sellers to be booked

Section 18, sub-section 2: If the seller commits any thing against the act, they shall be labile for fine ranging up to Rs 200,000.

9. Complaint response mechanism

Section 20: In case of inability to deliver the goods within the stated time,the customer can file the complaint in concerned District Administration Office.

Section 23: The seller should designate an employee to hear complaints from the customers. The name, phone number and address of the employee shall be placed on the website

E-commerce in Nepal

Ecommerce is still immature in Nepal. Even though the website like Kaymu.com.np, Meroshopping.com etc are doing well, they still lack the public exposure. Buying something from Internet is myth in Nepal. People still afraid to do it. Before a decade eCommerce was setup as sending gifts and money online and other websites promoting “Send Gifts to Nepal” which had merely a concept of eCommerce It was target to Nepali residing in USA, UK, Australia and Europe. There was no effect of that business to support the eCommerce concept in Nepal. Gradually the business was promoted by other companies who saw there was a marginal profit. Along with the rise of IT, and business concept many online stores were launched but they didn’t have the actual process of buying and selling online. They were the virtual stores with the best example which gave a concept of selling and buying online but not paying.

Many companies have started the trend of eCommerce in Nepal decade ago, but the challenges are still the same. The actual sales have not been able to start due to lack of knowledge, awareness and online payment systems. Selling globally and inside the country is the same in years. Payment Gateways are being developed but they have their limitation due to legal and security issues. The lack of proper knowledge and awareness among the generation is the major hurdle in eCommerce. In past recent years due to education and reach to internet among the many people in Nepal has brightened the future of eCommerce The interest of students in the field of IT, the growth of IT companies has helped a lot in the awareness and interest in young generation for internet and IT, has directly created more opportunities for the growth of eCommerce in Nepal.

Many online portals and shopping portals are launched now. Leaving the measurement of success behind, they are now on the top list. Peeping into the future of eCommerce, launch of few large online shopping portals was thought as milestone, everyone thought there will be a turnaround in the eCommerce industry in Nepal. Now having dozens of virtual Nepali stores in the web, they still have the same problem of payment and a belief of people, they still have a level of trust to build among the visitors.

1. <u>Here are the list of some ecommerce websites of Nepal.</u>
2. Kaymu.com.np
3. Nepbay.com.np
4. Meroshopping.com
5. Harilo.com
6. Foodmandu.com
7. Yeskantipur.com
8. Muncha.com
9. Metrotarkari.com
10. Rojeko.com
11. Estornepal.com
12. esewa.com
13. ticketsewa.com
14. hamrobazar.com
15. ipay.com.np
16. sabaikomart.com
17. metrotarkari.com
18. foodmandu.com
19. paybill.com
20. hellopaisa.com
21. karobarmart.com
22. sastodeal.com and many more

Growth of Commerce in Nepal

Now there are multiple eCommerce web portals and social platforms who are directly involved in online shopping. If u want to buy a new cloth, a book, a mobile phone or even order your lunch online from your home. Yes, it is possible in Nepal and there are a lot of companies offering these services. Before a decade eCommerce was setup as sending gifts and money online and other websites promoting “Send Gifts to Nepal” which had merely a concept of eCommerce. It was target to Nepali residing in USA, UK, Australia and Europe. There was no effect of that business to support the eCommerce concept in Nepal. Gradually the business was promoted by other companies who saw there was a marginal profit. Though problems like lack of proper payment gateway, transportation and internet reliability are existent in Nepal, many innovators have developed some innovative e-commerce services in Nepal. There are a list of some e-commerce sites in Nepal as presented above . Some of the popular e-commerce business in Nepal are:

Traditional e-commerce sites:

<u>Muncha.com:</u> Muncha is probably the most popular e-commerce site in Nepal. It is popular among non-residential Nepalis to send gifts to any part of Nepal. The site is popular for buying birthday cakes, flowers, gifts, toys as well as electronic items. You can pay via. E-banking, credit card or even cash-on delivery.

YesKantipur.com: Launched by Kantipur Publications, Yeskantipur aims to provide everything to your doorstep. A wide variety of items including books, electronic items, home appliances, food and beverages are available from different vendors. The services are available only inside Kathmandu valley with cash-on-delivery or e-banking. There is no delivery charge for items inside ring road. Special discounts are announced on the site in regular interval.

Rojeko.com: Rojeko has a wide variety of items for those willing to buy products inside Nepal or those willing to send gifts to Nepal. Rojeko is mostly focused on electronic items. Payment can be done via. Paypal, credit cards, e-sewa or cash-on-delivery. There is always a crazy offer with heavy discount for a product.

Bhatbhatenionline.com: The popular departmental store, Bhatbhateni has its own e-commerce site. Different types of items available on departmental store are also available on its online portal. Payment can be done via. Paypal, e-banking or credit card.

Some innovative e-commerce products:

Apart from traditional e-commerce sites, some entrepreneurs have implemented revolutionary ideas in Nepali market. With features of viewing and ordering products online, these services have their specialties which make them different from other sites. Here are a few innovative ideas implement in Nepal:

Harilo.com: Harilo allows people in Nepal to buy good from US retails and pay from Nepal. You can simply oder products from any US-based websites like Amazon and get them delivered to you in Nepal via. Harilo. In other words, you get the same services as US citizen. You can pay via. E-banking, Paypal and a lot of other options which also includes writing a check to Harilo.

MetroTarkari.com: Recently launched by a group of young enterpreneurs, Metrotarkari is a unique product in Nepal which delivers vegetables at your doorstep. With an ambition of reducing the gap between farmers and customers, the site allows users to select vegetables, fruits or meat online and get them delivered at the doorstep with cash-on-delivery facility.

Foodmandu.com: Foodmandu is another unique service, which delivers food from different restaurants in Kathmandu and Lalitpur to your doorstep. It has a list of 75 different restaurants in valley and their menus from which you order any items of your choice. The items get delivered within an hour and payment can be cash-on-delivery.

Here are some categorized eCommerce Business Practices and Trend in Nepal.

Practice of Send gifts: Send Gifts to Nepal is most oldest and successful online shopping business practice in Nepal till date. Many websites were started to promote the local sales via online, with the concept of send gifts to Nepal. Their target was Nepali people living in USA, Europe Australia and Canada and other countries abroad.These websites promoted the trend of sending gifts to Nepal and even included money transfer facilities. It was easy for those to send gifts to their family and loved ones. The price tag is little high but the service is worth paying. Website owners are more interested in send gifts trend and practice rather than local sales. We suggest this to be the best online shopping practice, but not locally.

Free Classified and online Market web portals: Free classified websites have been started and to give buyers a real experience in Online Shopping. In this trend sellers could promote their products and services. They enabled direct sellers and buyers interaction. This concept lacked the medium of online payment. In Nepal online payment system is a issue to discuss. In the classified practice of eCommerce business both used and New Products are allowed to sell. Its just a platform to showcase your products and services. Its more popular for second hand market over the Internet.

Showcasing over Internet/Online Shopping: This concept was easy for everyone to display their products in the website. If anyone orders the product they can choose to pay via bank transfer or pay cash on delivery. You can also pay via bank and credit cards. National local payment gateways are also used. Some of food portal have also a good concept of giving individuals to deliver their food online.

Get latest products from international shopping portals: There are few web portal , marking as a online shopping portal. They buy the product you choose and pay for them.In this trend of eCommerce, buyers can easily buy latest and desired goods from international market. Just you need to provide the product detail and pay them, They will deliver them at your doorstep.When you don't have a medium to pay online with credit card, then this is a best option for you. The website owners have a real shopping experience but buyers just get their goods delivered.

Social Media Selling Platforms: Its the latest and popular, online shopping and eCommerce experience in Nepal. Sellers just create page and promote their products. In some cases they don't have stocks and a shop, they are the actual online store , who deliver your products after you order with them. If you are interested you can order online and pay them at the time of delivery. Here you don't have the actual experience buying over the internet, but you can use social media to get your goods home. here sellers are making advantage of Facebook generation.

Over the past decade, e-commerce has transformed the way the business is being done in the developed world. It instantly became a successful model in regions like USA and Europe, and now the developing economies are also using this tool to grow their business. Nepal is a developing economy and according to The Economic Times, the country will experience a dip in economic growth in the year 2015 due to political turmoil and deadly earthquakes that disrupted the business activity in the country. However, the statistics compiled by the World Bank also suggest that there will be improvement in the year 2016 as the growth rate will climb up from 3.4 percent to 3.7 percent. This economic growth can be attributed to the flourishing e-commerce sector in the country.

With the influx of latest technology from neighboring countries, such as India, Nepal is on its way to success through e-commerce. Online shopping is taking the Nepalese market by storm and it is facilitated by fast speed inexpensive 3G and 4G internet technology. The convenient modes of payments and user friendly shopping apps are further paving a pathway to unprecedented growth in the e-commerce sector. According to a report compiled by Kaymu, a leading online marketplace in Nepal, the e-commerce sector is growing at a rapid pace, people are showing more trust in these marketplaces that offer a range of products from electronic items to home appliances to apparels. People are more inclined towards purchasing mobile phones, used motor vehicles, motorcycles, computer accessories and laptops through online platforms and this is evident from the pie chart below:

These e-commerce ventures are giving the traditional brick and mortar stores a run for their money. Some of the leading enterprises are now digitizing their operations and offer online shopping options via their official websites. The businesses now realize the significance of e-commerce and are now reserving a good chunk of their revenue to be spent on digital marketing. The advantage they aim to achieve is that not only they will have increased online presence, but can also get feedback from their consumers instantly via social media sites and other digital modes.

Compared to developed economies where the e-commerce industry has faced a boom and matured, developing economies such as Nepal have a lot of room for expansion and development. All the trends and signs point toward a positive direction. The graph below demonstrates that the majority of people in Nepal visit various e-commerce websites between 11: 00 am to 04: 00 pm and most of the transactions also take place during these hours.

The developing economies may have joined the bandwagon a little late, but they are also slowly realizing the potential and benefits that e-commerce sector offers. The e-commerce sector is still in its initial phase in Nepal, however many successful ventures, such as Kaymu and **Hamrobazar.com** are important actors playing a very crucial role in the growth of this industry.

Analysis of E-Business For Nepal

Strengths:

With the influx of latest technology from neighboring countries, such as India, Nepal is on its way to success through e-commerce

Many online portals and shopping portals above 20 are launched now.

Also getting feedback from consumers instantly via social media sites and other digital modes has been easier to improve the service and product

People are showing more trust in these marketplaces that offer a range of products from electronic items to home appliances to apparels.

The businesses now realize the significance of e-commerce and are now reserving a good chunk of their revenue to be spent on digital marketing. The advantage they aim to achieve is that not only they will have increased online presence, but can also get feedback from their consumers instantly via social media sites and other digital modes.

Weaknesses :

Government tax policies and cross-border customs regulations,

Unavailability of adequate and secure payment solutions,

Shipping and handling standards and Internet access costs.

Most people don't own credit cards, these companies have found a way to not only find a niche but also entice more people to opt for shopping online.

Cash on delivery becomes the other problem with a lot of cases of goods being returned

The average age of customers buying online is 21 years and the business is mainly Kathmandu centric

There is no department to take responsibility for damage or theft in the process of delivery

Online business like thamel.com is seasonal in Nepal and is not being able to scale up as it should due to the lack of local supply chain

Very few banks in Nepal have a proper system of online payment. Less than five per cent of Nepali buyers as of today use online payments. "A lot of work has been going on to connect all the banks so that interbank transfer across the border will be easier

Opportunities:

The market size of e-commerce in Nepal is USD 75.4 million and is multiplying three times every year.

Improvement in the Internet infrastructure and government regulations, which would create cheaper and more reliable online services and better payment gateways

Smartphones have penetrated the market in Nepal, giving rise to changing market trends and customer expectations. This wireless technology has enhanced ecommerce capabilities and changed the way how people do business today

Compared to developed economies where the e-commerce industry has faced a boom and matured, developing economies such as Nepal have a lot of room for expansion and development.

Threats:

Offline purchasing in Nepal still dominating the market

The market is fragmented and involves a lot of small players.

Challenges of service providers is the longer payment cycle where small sellers suffer the most

Online markets in Nepal are not competing with each other but trying to make a secured place for themselves

Considering Nepal to be a high risk zone, Nepali cards were not allowed to be used in e-commerce in other countries until a couple of years ago

Problems of ecommerce in Nepal

Online Shopping, clearly states that you must be able to buy online and pay, get the goods delivered. If not, we cannot call it a online shopping platform. To give a complete online shopping experience every shopping portals and business must provide a minimum of online payment and delivery system. Every consumer must be able to get their products delivered in a complete shopping solution

In-context to Nepal, online payment limitation is the major hurdle in eCommerce business. Now having dozens of virtual Nepali stores in the web, they still have the same problem of payment and a belief of people, they still have a level of trust to build among the visitors. Nepali government doesn’t allow us to sell anything that’s digital. And there are restrictions on mobile phones and wireless routers or small things like camouflage prints or multi-vitamin tablets. This not only restricts our customer base but the government also misses out on the tax payments they could have received .Hope we will get the solution soon.

Future of E commerce in Nepal

It’s a practice which no one can escape and avoid. It will be either today or tomorrow we must adapt to it. Both the development of payment system, legalizing and securing the payments, creating reliability from the portals is a major issue. With the growing generation every business will have to adapt and think about their online existence in future depending on the nature of business. It will not be too long; there will be further boom in the trend of eCommerce business soon in Nepal.

Recommendation for development of E-commerce business in Nepal

Following recommendation can be given for the problem stated above:

Well developed infrastructure, investment from various players and logistic providers should be there in order to increase the number of buyers and the business on the whole. “With the increase in the number of the customers and the purchase, the cost of delivery will automatically minimize.

The offline business chain, the online business should also build their own network which helps in the growth of sales

Interbank transfer should be made easier and that customers should be encouraged to make online payments. Likewise building up the easy ways to return the goods could help both buyers and sellers. As far as trust issue is concerned, he believed that it can be solved together by increasing awareness and ensuring quality of products.

Public awareness must be created regarding the pros and cons of ebusiness

Government flexible and helpful policies must be introduced

Online payment through banking must be increased

The ebusiness should focus every nook and corner of the country

Availability of adequate and secure payment solutions should be made

Conclusion

We must create history not become a history. If business persons are always active on the development, process and the business policy and strategy on eCommerce business for their clients surely E-business will Boom.However, E-business should offer different online payment system verification. Payments should be easy and safer. Every individual’s trust must be built by business owners. Time has come to take ones business online compulsorily in this IT era, so if you are thinking of being a successful businessperson start your online shop today!!! But you must be aware of the problems and prospect of it and be aware of threats and weaknesses and you should be able to fully take advantages of your strengths.

Assignments

1.Define E-Commerce.Write the differences between traditional commerce and e-commerce.

2.Explain the different components of E-Commerce.

3.Explain the different types of E-Commerce.

4.What are the applications of E-Commerce ? Explain.

5.Compare the advantages and disadvantages of E-Commerce.

6.Explain the advantages and disadvantages of E-Commerce.

7.Explain the situation and future scope of E-Commerce in Nepal.

8.What are the scopes of E-Commerce in Nepal.

9.Explain the difficulties of E-Commerce in Nepal.Write your own opinion.

Unit-3:E-governance

Contents

3.1:Introduction to E-governance

a. Electronic Governance is the application of Information and Communication Technologies (ICTs) for delivering government services through integration of various stand-alone systems between Government-to-Citizens (G2C), Government-to-Business (G2B), and Government-to-Government(G2G) services. It is often linked with back office processes and interactions within the entire government framework. Through **e-Governance**, the government services are made available to the citizens in a convenient, efficient, and transparent manner.

b. Electronic governance or e-governance can be defined as the usage of Information and Communication Technology (ICT) by the government to provide and facilitate government services, exchange of information, communication transactions and integration of various stand-alone systems and services.

c. In other words, it is the use of technology to perform government activities and achieve the objectives of governance. Through e-governance, government services are made available to citizens and businesses in a convenient, efficient and transparent manner. Examples of e-governance include Digital India initiative, National Portal of India, Prime Minister of India portal, Aadhaar, filing and payment of taxes online, digital land management systems, Common Entrance Test etc.

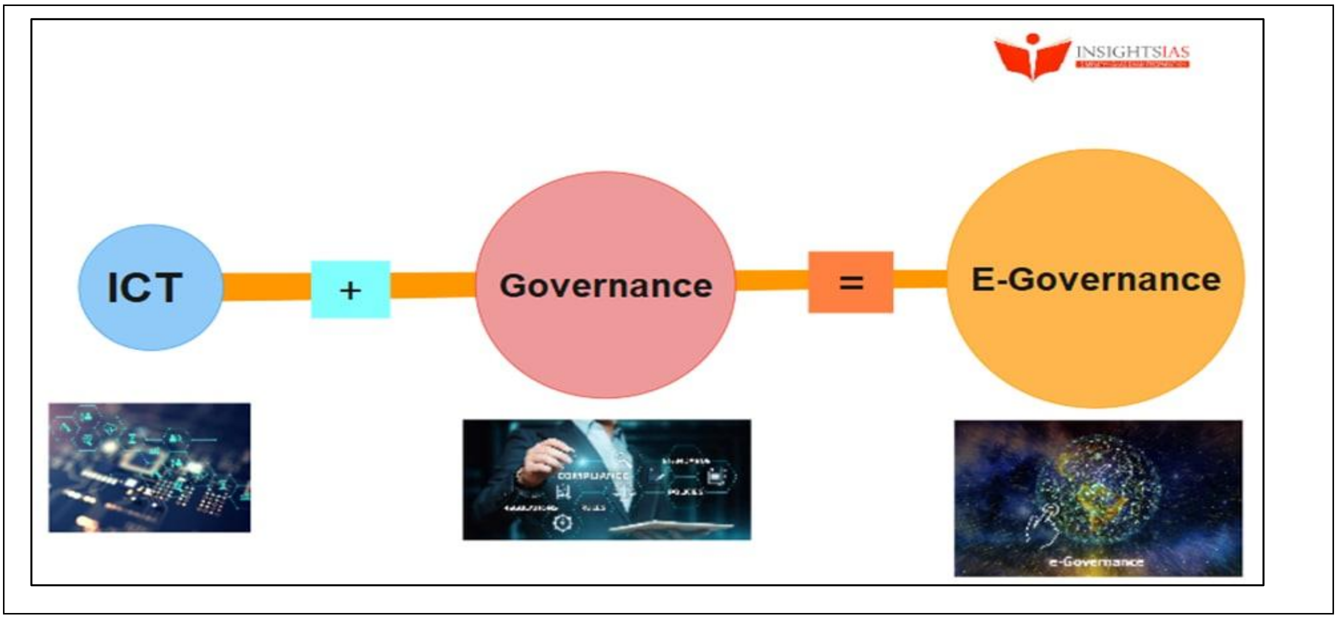
d. EGov is all about the delivery of national or local government information and services within the reach of the internet or other digital or social media to the hands of the citizens, businesses and other governmental organizations.

e. **According to the World Bank (2000)** – EGov is the all about the use of ICT services by government agencies in order to promote associations between citizens, businesses, and other arms of the government. This serves as a tool to many aspects including better delivery of government services to citizens, improved communications with businesses and citizen empowerment made available through the ease of information available along with proficient management. The consequential benefits are decreased bribery, better transparency, more accessibility, growing revenue, and reduced cost."

f. Through the means of e-governance, government services are made available to citizens in a suitable, systematic, and transparent mode. The three main selected groups that can be discriminated against in governance concepts are government, common people, and business groups.

g. E-governance is the best utilization of information and communication technologies to mutate and upgrade the coherence, productivity, efficiency, transparency, and liability of informational and transnational interchanges within government, between government agencies at different levels, citizens & businesses.

h. The UNESCO states that E-governance is the public sector’s use of information and communication automation in order to upgrade information and service delivery, stimulating resident involvement in the decision-making process and making government more liable, unambiguous and productive



Objectives of e-Governance

The objectives of e-governance can be listed down as given below:

a. To support and simplify governance for government, citizens, and businesses.

b. To make government administration more transparent and accountable while addressing the society’s needs and expectations through efficient public services and effective interaction between the people, businesses, and government.

c. To reduce corruption in the government.

d. To ensure speedy administration of services and information.

3.2 Components of E-Governance

The three components of E-Governance are:

1.Government

Government is the organization which takes decisions and makes laws for the citizens of a country. The political system by which a country or community is administered and regulated.

2.Citizens

A citizen is a person who, by place of birth, nationality of one or both parents, or naturalization is granted full rights and responsibilities as a member of a nation or political community.

3.Businesses

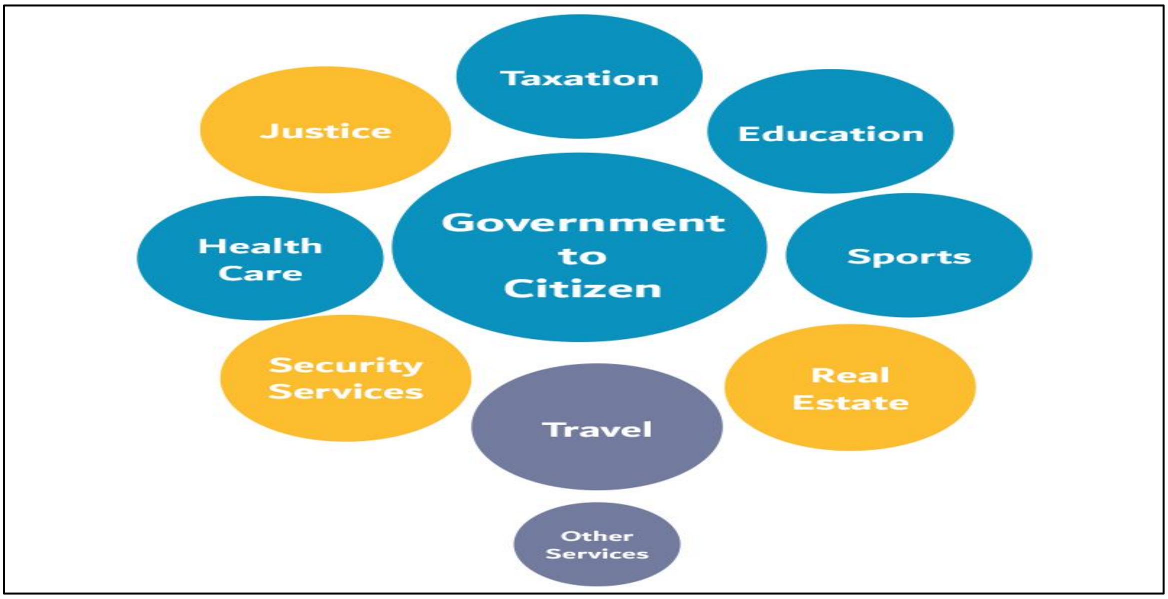
Business means an organization or enterprising entity engaged in commercial, industrial, or professional activities.

Types of E-Governance

There are 4 types of E-Governance which are listed with description below:-

Government-to-Citizen (G2C)

- The Government-to-citizen refers to the government services that are accessed by the familiar people. And Most of the government services fall under G2C. Likewise, the primary goal of Government-to-citizen is to provide facilities to the citizen. It helps the ordinary people to reduce the time and cost to conduct a transaction. A citizen can have access to the services anytime from anywhere.
- Furthermore, Many services like license renewals,paying electricitybill and paying tax are the some example in G2C. Likewise, spending the administrative fee online is also possible due to G2C. The facility of Government-to-Citizen enables the ordinary citizen to overcome time limitation. It also focuses on geographic land barriers.
- Information is exchanged within the government i.e., either, between the central government, state government and local governments or between different branches of the same government.
- The citizens have a platform through which they can interact with the government and get access to the variety of public services offered by the Government.



Government-to-Business (G2B)

- Government-to-business (G2B) is a relationship between businesses and government, where government agencies of various levels provide services or information to a business entity via government portals or with the help of other IT solutions.
- Such assistance from the government bodies' side aims to facilitate swift and smooth business operations, as well as a fair and transparent environment to do business.

The range of the G2B services is broad.Some of them are listed below:

a. Online information and advisory services
b. Government contracting
c. Digital procurement marketplaces
d. Business licenses, permits, and regulation updates
e. Electronic auctions
f. Tax, social insurance payments, and reporting
g. Electronic forms
h. Online application submission functionalities
i. Virtual business dispute resolution
j. Online startup registering

Government-to-Government(G2G)

- Government to government (G2G) is the electronic sharing of data and/or information systems between government agencies, departments or organizations. The goal of G2G is to support e-government initiatives by improving communication, data access and data sharing.

- Information is exchanged within the government i.e., either, between the central government, state government and local governments or between different branches of the same government.
- The Government-to-Government refers to the interaction between different government department, organizations, and agencies. This increases the efficiency of government processes. In G2G, government agencies can share the same database using online communication. The government departments can work together. This service can increase international diplomacy and relations.
- In conclusion, G2G services can be at the local level or the international level. It can communicate with global government and local government as well. Likewise, it provides safe and secure inter-relationship between domestic or foreign government. G2G constructs a universal database for all member states to enhance service.

Government-to-Employee

- A method of providing government and government employee interaction in order to transact transactions such as payroll and pension planning, obtain training opportunities, and apply to public benefits.
- Using government-to-employees (abbreviated G2E) allows governments and their employees to communicate via instantaneous communication methods online. To achieve electronic government, G2E represents one of the four primary models.
- In government circles, G2G interaction refers to the exchange of information and services. As one of the biggest employers in any economy, the government deals with its workers as if other employers do, with its G2E (government-to-employee) programme.
- G2E provides online facilities to the employees. Likewise, applying for leave, reviewing salary payment record. And checking the balance of holiday. The G2E sector provides human resource training and development. So, G2E is also the relationship between employees, government institutions, and their management.

3.3:Advantages of E-Governance

a. **Speed**: Technology makes communication fast. Internet, smartphones have enables instant transmission of high volumes of data all over the world.E-governance helps the government to provide services fast.

b. **Saving Costs:** A lot of Government expenditure goes towards the cost of buying stationery for official purposes. Letters and written records consume a lot of stationery. However, replacing them with smartphones and the internet can saves crores of money in expenses every year.

c. **Transparency**: The use of e-governance helps make all functions of the business transparent. All Governmental information can be uploaded onto the internet. The citizens access specifically access whichever information they want, whenever they want it, at their convenience.However for this to work the Government has to ensure that all data as to be made public and uploaded to the Government information forums on the internet.Transparency control the corruption.

d. **Accountability**: Transparency directly links to accountability. Once the functions of the government are available, we can hold them accountable for their actions.

Disadvantages of E-Govenance

a. **Loss of Interpersonal Communication:** The main disadvantage of e-governance is the loss of interpersonal communication. Interpersonal communication is an aspect of communication that many people consider vital.

b. **High Setup Cost and Technical Difficulties**: Technology has its disadvantages as well. Specifically, the setup cost is very high and the machines have to be regularly maintained. Often, computers and internet can also break down and put a dent in governmental work and services.

c. **Illiteracy**: A large number of people in Nepal are tech-illiterate and do not know how to operate computers and smartphones. E-governance is very difficult for them to access and understand.

d. **Cybercrime/Leakage of Personal Information**: There is always the risk of private data of citizens stored in government serves being stolen. Cybercrime is a serious issue, a breach of data can make the public lose confidence in the Government's ability to govern the people.

Problems for implementing E-governance in Nepal

The EGov Index of Nepal shows its low ranking compared to other neighboring countries. Nepal has many challenges for the implementation of EGov systems, which starts from the low literacy rate to the lack of infrastructure and human resources .

While e-governance provides the advantages of convenience, efficiency and transparency, it also has problems associated with it. They are as follows:

- Lack of computer literacy: Nepal is still a developing country and a vast majority of the citizens lack computer literacy which hinders the effectiveness of e-governance.
- Lack of accessibility to the internet or even computers in some parts of the country is a disadvantage to e-governance.

- e-Governance results in a loss of human interaction. As the system becomes more mechanised, lesser interaction takes place among people.
- It gives rise to the risk of personal data theft and leakage.
- e-Governance leads to a lax administration. The service provider can easily provide excuses for not providing the service on technical grounds such as “server is down” or “internet is not working”, etc.

E-Governance in the Indian context

e-Governance in India is a recently developed concept. The launch of National Satellite-Based Computer Network (NICENET) in 1987 and subsequent launch of the District Information System of the National Informatics Centre (DISNIC) programme to computerise all district offices in the country for which free hardware and software was offered to the State Governments provided the requisite impetus for e-governance.

e-Governance there after developed with the growth of technology. Today, there are a large number of e-Governance initiatives, both at the Union and State levels. In 2006, the National e-Governance Plan (NeGP) was formulated by the Department of Electronics and Information Technology and Department of Administrative Reforms and Public Grievances that aims at making all government services accessible to the common man, ensure efficiency, transparency and reliability of such services at affordable costs to realise the basic needs of the common man.

The NeGP has enabled many e-governance initiatives like:

1.**Digital India** was launched in 2015 to empower the country digitally. Its main components are:

- Developing a secure and stable digital infrastructure
- Delivering government services digitally
- Achieving universal digital literacy

2.Aadhaar is a unique identification number issued by UIDAI that serves as proof of identity and address on the basis of biometric data. It is being used to provide many benefits to the members of the society. One can **e-sign** documents using Aadhar.

3.myGov.in is a national citizen engagement platform where people can share ideas and be involved with matters of policy and governance.

4.UMANG is a Unified Mobile Application which provides access to central and state government services including Aadhar, Digital Locker, PAN, Employee Provident Fund services, etc.

5.Digital Locker helps citizens digitally store important documents like mark sheets, PAN, Aadhar, and degree certificates. This reduces the need for physical documents and facilitates easy sharing of documents.

6.PayGov facilitates online payments to all public and private banks.

7.Mobile Seva aims at providing government services through mobile phones and tablets. The m-App store has over 200 live applications which can be used to access various government services.

8.Computerisation of Land Records ensures that landowners get digital and updated copies of documents relating to their property.

The history of e-Governance activities in Nepal

The first IT policy of the GoN was drafted in 2000. The IT policy 2000 underwent numerous amendments and changes; coming up to the IT Policy 2015. The major problem with the implementation of IT policy 2000 was majorly the political and social instability (Lawoti, 2003). Several infrastructures and institutions namely HLCIT, NITC, MoEST, MoIC, MoGA, and MoF initiated the eGMP. The report was then prepared by KIPA (KIPA, 2006). The ETA 2008, which was considered the outstanding success in IT regulation in Nepal was crucial for the implementation of EG. The right to information act, 2007 also supported implementation of EG (Karki, 2007). The establishment of GIDC under NITC was possible with the assistance of KOICA (KIPA, 2006). The construction of IT Park in Banepa close to the capital became additional institution formed under this policy however the plan was not so much successful (MartinChautari, 2014). The government-owned NTA later privatized and changed into NTC and named Nepal Telecom (Gautam, 2016). The IT policy of 2010 was enacted following the failure of IT policy 2004. Further to the eGMP, the draft of the wireless broadband master plan 2012 was prepared by ITU forthe powerful use of broadband generation in Nepal and thatstands today as the strongest strategic planning of EG in Nepal (ITU, 2012). Until date, there are lots of policies, acts and regulations being published in Nepal for the betterment of ICT services, however, the IT policy of 2015 is still latest policy in implementation. The policy was proposed to withstand the foundation for the vision of Digital Nepal Volume 1, Issue 2 (December 2019) ISSN: 2705-4683; e-ISSN: 2705-4748 LBEF Research Journal of Science, Technology and Management 24 (Ministry of Information and Communications, 2015).

The current situation of e-Governance in Nepal

Nepal IT Policy 2000 aims to interlink all ministries, departments, and offices with the GoN with internet to provide services online. The eGMP Consulting Report prepared by KIPA was figured alongside the most feasible government projects. The mission of the report was to provide value added quality service through ICT. (KIPA, 2006) In addition, the policy highlighted the use of ecommerce and distant education with having EG at the facilitator role for the government. ADB reinforced “transformation” program that has prioritized twenty-two services including NID, driving-license, LRMS and rural e-community (Illawara Technology, 2007). In the words of Bekkers, EG is the major action plan to improve Nepal’s fragile government. Since the outcomes on the use of ICTs in public organizations are specific and situation-dependent, there is a significant need for recording evidence of some pilot EG program implementations for future reference. (Bekkers, 2007). The availability of updated and trustworthy information in the government websites (Rani and Kautish, 2018) (Kaur and & Kautish, 2019) are lagging a lot. The innovative strategies in the EG projects hence ensure that the government services are not impacted by poor delivery services, this is done by equipping the surveillance devices over resident's personal and private information. (Thomas, 2012) The citizen-oriented data made available on the internet and activities made available on the smartphones have accumulated a heavy amount of data in unstructured form. Hence, Nepal EGov services can initiate using the Big Data and Data Mining technologies to resolve many governmental services that are getting problems due to data inconsistencies. (Shakya, 2018).

Interactive-Service model of e-Governance

This model also called the government-citizen-government model (G2C2G) is the framework developed by scientists whereby the government provides services to its citizens with having a proper consultant with them about the new services. The government takes upon public opinion before launching the new services for their citizens. This is moreover a win-win situation created by the combined decision of both the parties. Applications of G2C2G Submission of grievances and queries by the public to the government• Establishment of the comprehensive communication channels for support• Used in carrying out transactions into the online platform for the provision of filing taxes and9• Volume 1, Issue 2 (December 2019) ISSN: 2705-4683; e-ISSN: 2705-4748 LBEF Research Journal of Science, Technology and Management 25 payment

Major challenges with e-Governance implementation in Nepal

Nepal is one of the fast-growing developing countries in Asia with having the latest ICT image of the first 4G service hosting Asian country in the world. Despite having abundant growth in the use of IT and mobile devices, the country remains behind in the light of darkness while it comes to its development in front of western countries; this is strict because of the low literacy rate of the country. The per capita is too low and the poverty is still pulling back the development of the nation. The ever fluctuating political instability, corruption and tough geographical vegetation of the country standstill causing the EG implementation in Nepal to face intense challenge. Nepal is yet to develop its infrastructure and ability to incorporate full-fledged EG. Below listed are some of the major challenges for EG implementation in Nepal: 1. Literacy The literacy rate of Nepal was only around 64.66% in 2015 ("Nepal | UNESCO UIS," n.d.) which is very low in comparing to its neighboring countries. With the presence of multiple languages and religion in the country, there also exist language issues. English cannot be considered the third language as most of the population even could not speak and understand the National language due to illiteracy. (Chapagain, 2006). 2. Lack of human resources Today the national economy of Nepal is persistent with the remittance of Nepali migrant workers staying and working abroad in the Gulf countries. The education system is not so good and assurance of getting good opportunity is very low in the country, which has affected the flow of competent human resources outside the country. This has ultimately created the situation of human resource lagging in the multiple fields of business. On the other hand, the government personnel selected nationally lags the minimal computer knowledge requirements leading into their reluctance towards use of technology. 3. Political Uncertainty Political uncertainty is the biggest challenge for Nepal. The undeterminable changes in the country's political representation have negatively impacted many infrastructure developments projects in the past and still today the same problem persists uniformly as the situation never improved in the last decade. The implementation of EG is still lacking which is evident from the case of NID card itself as the project is getting extended from last few years and despite the distribution in some part of the country, the project is still in the phase of public acceptance. 4. Lack of Coordination The lack of coordination and senior guidance is one of the biggest challenges ever in Nepal. (Kharel, 2012). There are very few leaders in Nepal who remained certain with their views and deeds, the country has a lot of political influence that resulted in low participation of the experts in coordinating activities. Therefore, the Implementation of EG requires strong leadership without which the implementation is impossible. 5. Weak Infrastructure Nepal has the most dangerous roads connecting the mountains, hills and terai. Due to the geographical imbalance of landscape and hardship in transporting the goods around the country, the minimal infrastructure requirement is also not fulfilled. Most of the government organizations still are using outdated hardware and equipment to accomplish their daily transactions. The lack of telecommunication infrastructures also stands as the major challenge for the implementation of EG. In Nepal, most of the government organizations are still using poor and outdated network devices and equipment. According to eGMP, the government organizations must develop the infrastructures in order to achieve the goal, vision, and objectives of EGov. (Purusottam Kharel, 2013) 2.5.1 Barriers associated with e-Governance implementation in Nepal

Unit-4: Cloud Computing and Internet of Things (IOT)

Contents

4.1:Introduction of Cloud Computing

- **Cloud computing is the delivery of different services through the Internet. These resources include tools and applications like data storage, servers, databases, networking, and software.**
- Cloud computing is an evolved branch of technology that helps businesses to meet their IT needs. This technical service helps both individuals and corporations daily. Everyone is dependent on good data storage facilities to overcome the storage limitations of hardware devices/systems. Everyone needs a pool of resources that is accessible anywhere, anytime.
- This is where cloud computing enters. It offers a very convenient and easy choice for clients to store data within a secured infrastructure. Therefore, many companies now rely on sound cloud computing solutions.
- Examples: Google Drive, Drop Box etc.

4.2 : Features & Components of Cloud Computing

Features of Cloud Computers

1. Self-service On-Demand

This is one of the most essential and significant characteristics of cloud computing. This means that cloud computing enables clients to regularly monitor the abilities, allotted network storage, and server uptime. Therefore, it is one of the most fundamental features of cloud computing that helps clients control various computing abilities as per their requirements.

2. Resources Pooling

This is also a fundamental characteristic of cloud computing. Pooling resources means that a cloud service provider can distribute resources for more than one client and provide them with different services according to their needs. Resource Pooling is a multi-client plan useful for data storing, bandwidth services and data processing services. The provider administers the data stored in real-time without conflicting with the client's need for data.

3. Easy Maintenance

This is one of the best cloud characteristics. Cloud servers are easy to maintain with low to almost zero downtime. Cloud Computing powered resources undergo several updates frequently to optimize their capabilities and potential. The updates are more viable with the devices and perform quicker than the previous versions.

4. Economical

This kind of service is economical as it efficiently reduces IT costs and data storage expenditure. Moreover, most cloud computing services are free. Even if there are paid plans, it's only to expand storage capacity, and these costs are often very nominal. This is a massive advantage of using cloud computing services.

5. Rapid Elasticity and Scalability

The best part of using cloud storage is that it can easily handle all the workload and data load concerning storage. Furthermore, as it is fully automated, businesses and organizations can save heavily on manual labour and technical staffing as cloud services are elastic, scalable and automated. This is one of the significant advantages of using cloud services.

6. Efficient Reporting services

The best aspect of using cloud storage is that even if it is fully automated and managed by bots, it has a speedy and prompt reporting service in case of any error or hindrance. In addition, the back-end cloud services team is pretty fast in addressing user reports, whether billing issues or functionality.

7. Automation

Automation is a pivotal characteristic of cloud computing. Cloud services can automatically configure, install and reboot to give optimum functionality. It reduces manual effort and is very user friendly. How commands are deployed through the cloud infrastructure requires no human interference concerning sorting the stored data.

8. Security

One of The best features of cloud computing is that it's secure. Clients hardly face any threat of piracy or data breach concerning the content uploaded overcloud. There is no threat of Trojans or viruses damaging the stored data. Even if the device happens to be infected, cloud data is safe from such harm.

9. Huge Network Access

A significant part of cloud services is that it's ubiquitous. Clients need any device with an internet connection to access cloud storage. In addition, the cloud service providers have ample access to the network that makes it easy for them to administer all the data uploaded on the Cloud through parameters like access time, latency, data output, and more.

10. Resilience

Resilience refers to the capability of the cloud service to recover quickly from any disruption. It depends on the speed of the internet and cloud database, servers and how quickly the network system recovers and reboots any damage or harm. Resilience also means that there are no physical barriers to accessibility. Clients can access cloud services remotely, making geographical nexus no bar for cloud usage.

Components of cloud computing

1.Network

Cloud resources are typically delivered to users over the internet, so there is a need for third-party service providers to build and maintain the networking infrastructure that makes this possible. This infrastructure includes physical wiring, switches, load balancers and routers that help ensure cloud infrastructure is always available for customers when needed.

2.Servers

A server is simply a computer or device that has been programmed to provide a service to a customer or user. There are web servers that serve HTML or PHP files using the HTTP protocol, file servers that store large volumes of information, mail servers that send e-mail over the internet and several other types. In private cloud deployments, organizations may use dedicated servers to store information, while public cloud providers use the multi-tenant model and may use the same server to provide services for more than one customer.

3.Storage

Cloud storage services allow organizations to store and manage data on off-site file servers instead of building their own physical data centers. Third-party cloud storage providers like Amazon Simple Storage Service (S3), Microsoft Azure and Google Cloud Storage can manage and maintain data along with remote back-ups. Data that is stored in the cloud can be accessed via the internet or queried by other applications that are deployed in the cloud. Cold data—data that is not actively being used—can also be stored in cloud infrastructure.

4.Virtualization

Virtualization may be the most important aspect of cloud infrastructure. Virtualization software abstracts the available data storage and computing power away from the actual hardware, enabling users to interact with their cloud infrastructure through a graphical user interface. Computing resources and data storage are often virtualized in cloud computing, making it easier for users to leverage these resources with added simplicity and less waste.

4.3: Classification of cloud computing

Classification of cloud computing

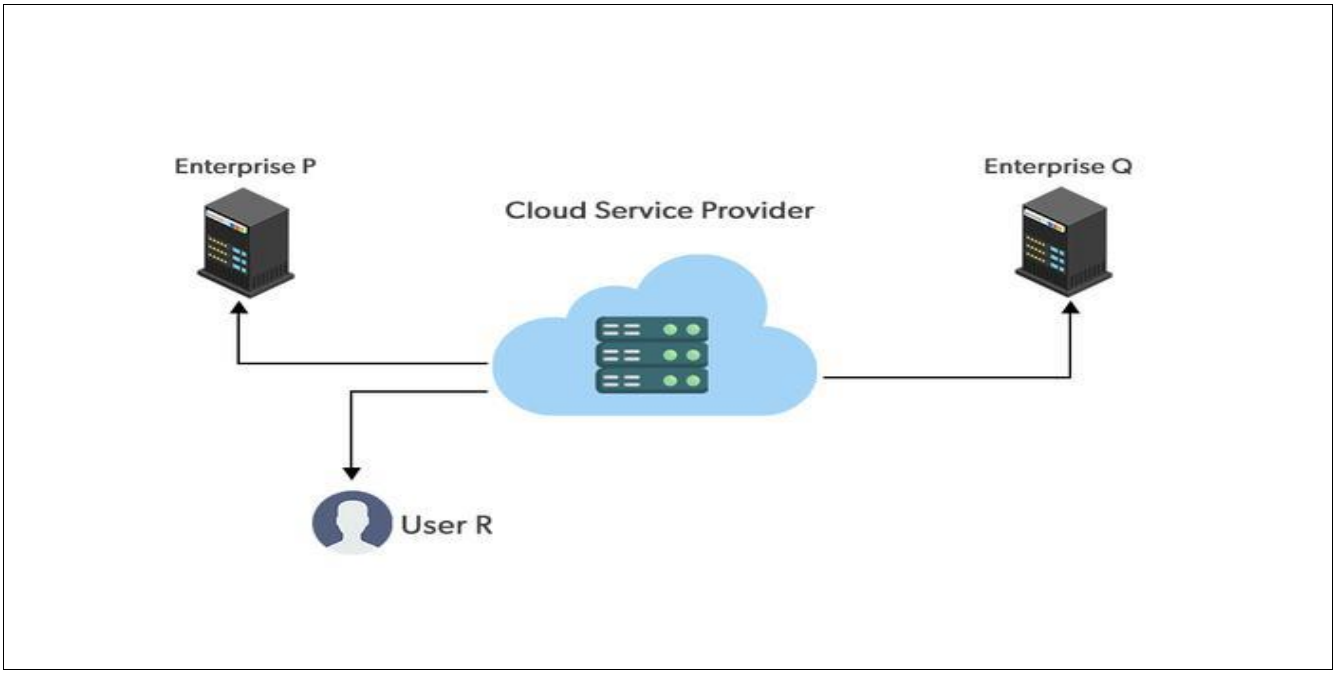
Cloud computing is Internet-based computing in which a shared pool of resources is available over broad network access, these resources can be provisioned or released with minimum management efforts and service provider interaction.

Types of Cloud

1. Public cloud
2. Private cloud
3. Hybrid cloud
4. Community cloud

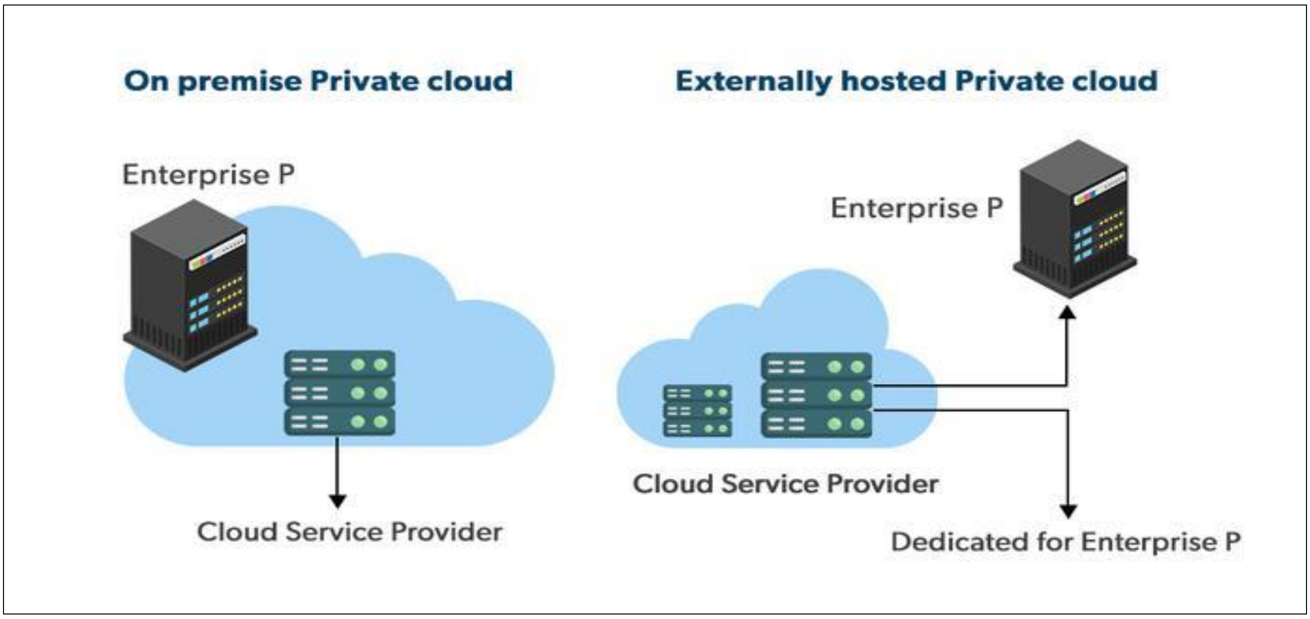
Public Cloud

- Public clouds are managed by third parties which provide cloud services over the internet to the public, these services are available as pay-as-you-go billing models.
- They offer solutions for minimizing IT infrastructure costs and become a good option for handling peak loads on the local infrastructure.
- Public clouds are the go-to option for small enterprises, which can start their businesses without large upfront investments by completely relying on public infrastructure for their IT needs.
- The fundamental characteristics of public clouds are **multitenancy**. A public cloud is meant to serve multiple users, not a single customer. A user requires a virtual computing environment that is separated, and most likely isolated, from other users.



Private cloud

- ❖ Private clouds are distributed systems that work on private infrastructure and provide the users with dynamic provisioning of computing resources.
- ❖ Instead of a pay-as-you-go model in private clouds, there could be other schemes that manage the usage of the cloud and proportionally billing of the different departments or sections of an enterprise.
- ❖ Private cloud providers are HP Data Centers, Ubuntu, Elastic-Private cloud, Microsoft, etc.



The advantages of using a private cloud are as follows:

a. **Customer information protection:** In the private cloud security concerns are less since customer data and other sensitive information do not flow out of private infrastructure.

b. **Infrastructure ensuring SLAs:** Private cloud provides specific operations such as appropriate clustering, data replication, system monitoring, and maintenance, disaster recovery, and other uptime services.

c. **Compliance with standard procedures and operations:** Specific procedures have to be put in place when deploying and executing applications according to third-party compliance standards. This is not possible in the case of the public cloud.

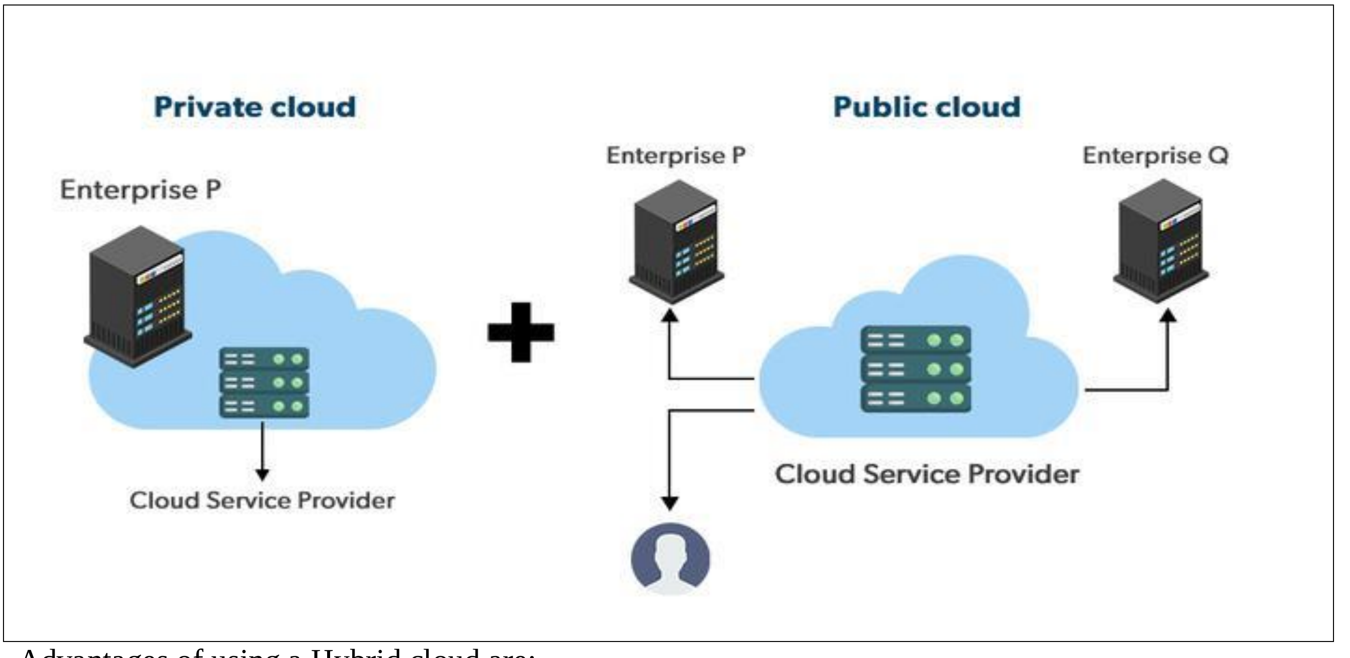
Disadvantages of using a private cloud are:

a. **The restricted area of operations**: Private cloud is accessible within a particular area. So the area of accessibility is restricted.

b. **Expertise requires**: In the private cloud security concerns are less since customer data and other sensitive information do not flow out of private infrastructure. Hence skilled people are required to manage & operate cloud services.

Hybrid cloud:

A hybrid cloud is a heterogeneous distributed system formed by combining facilities of the public cloud and private cloud. For this reason, they are also called **heterogeneous clouds.** A major drawback of private deployments is the inability to scale on-demand and efficiently address peak loads. Here public clouds are needed. Hence, a hybrid cloud takes advantage of both public and private clouds.



Advantages of using a Hybrid cloud are:

1) Cost : Available in cheap cost than other clouds because it is formed by distributed system.

2) Speed : It is efficiently fast with lower cost, It reduces latency of data transfer process.

3) Security : Most important thing is security. Hybrid cloud are totally safe and secured because it works on distributed system network.

Community cloud

- Community clouds are distributed systems created by integrating the services of different clouds to address the specific needs of an industry, a community, or a business sector. But sharing responsibilities among the organizations is difficult.
- In the community cloud, the infrastructure is shared between organizations that have shared concerns or tasks. The cloud may be managed by an organization or a third party.

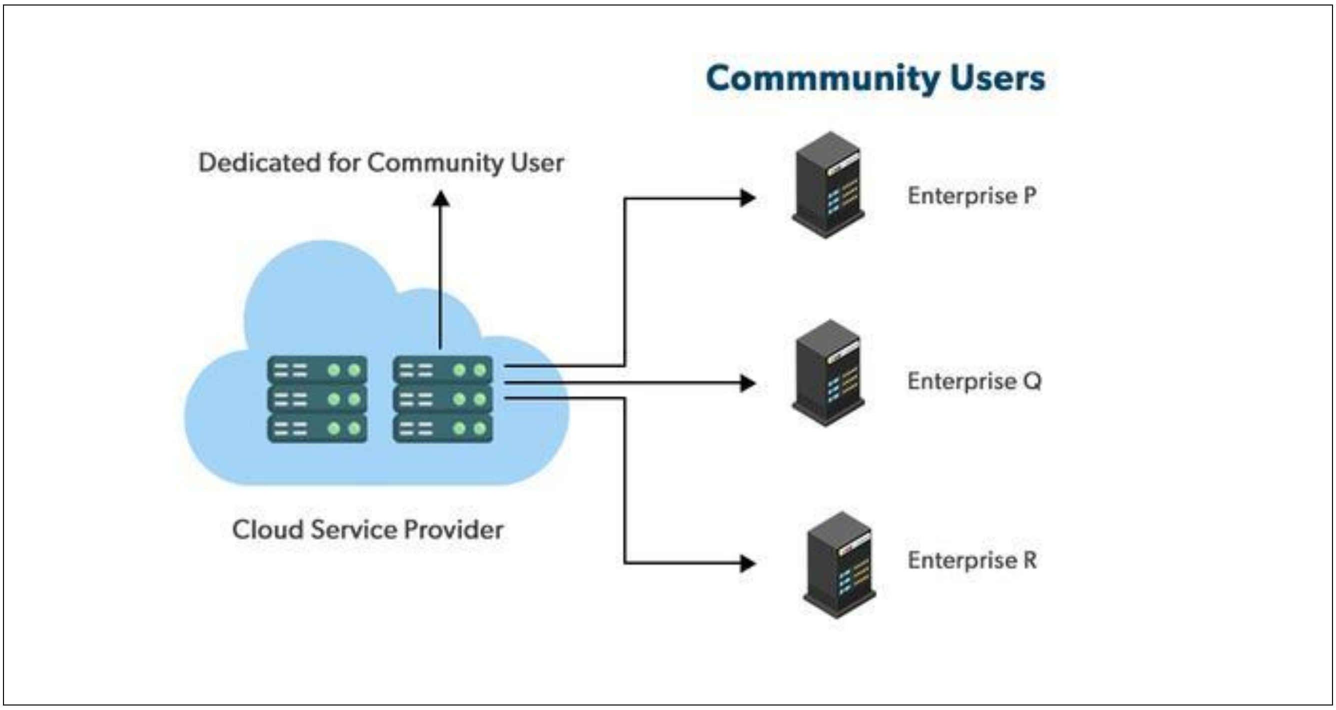
Sectors that use community clouds are:

1. Media industry: Media companies are looking for quick, simple, low-cost ways for increasing the efficiency of content generation. Most media productions involve an extended ecosystem of partners. In particular, the creation of digital content is the outcome of a collaborative process that includes the movement of large data, massive compute-intensive rendering tasks, and complex workflow executions.

2. Healthcare industry: In the healthcare industry community clouds are used to share information and knowledge on the global level with sensitive data in the private infrastructure.

3. Energy and core industry: In these sectors, the community cloud is used to cluster a set of solution which collectively addresses the management, deployment, and orchestration of services and operations.

4. Scientific research: In this organization with common interests in science share a large distributed infrastructure for scientific computing.



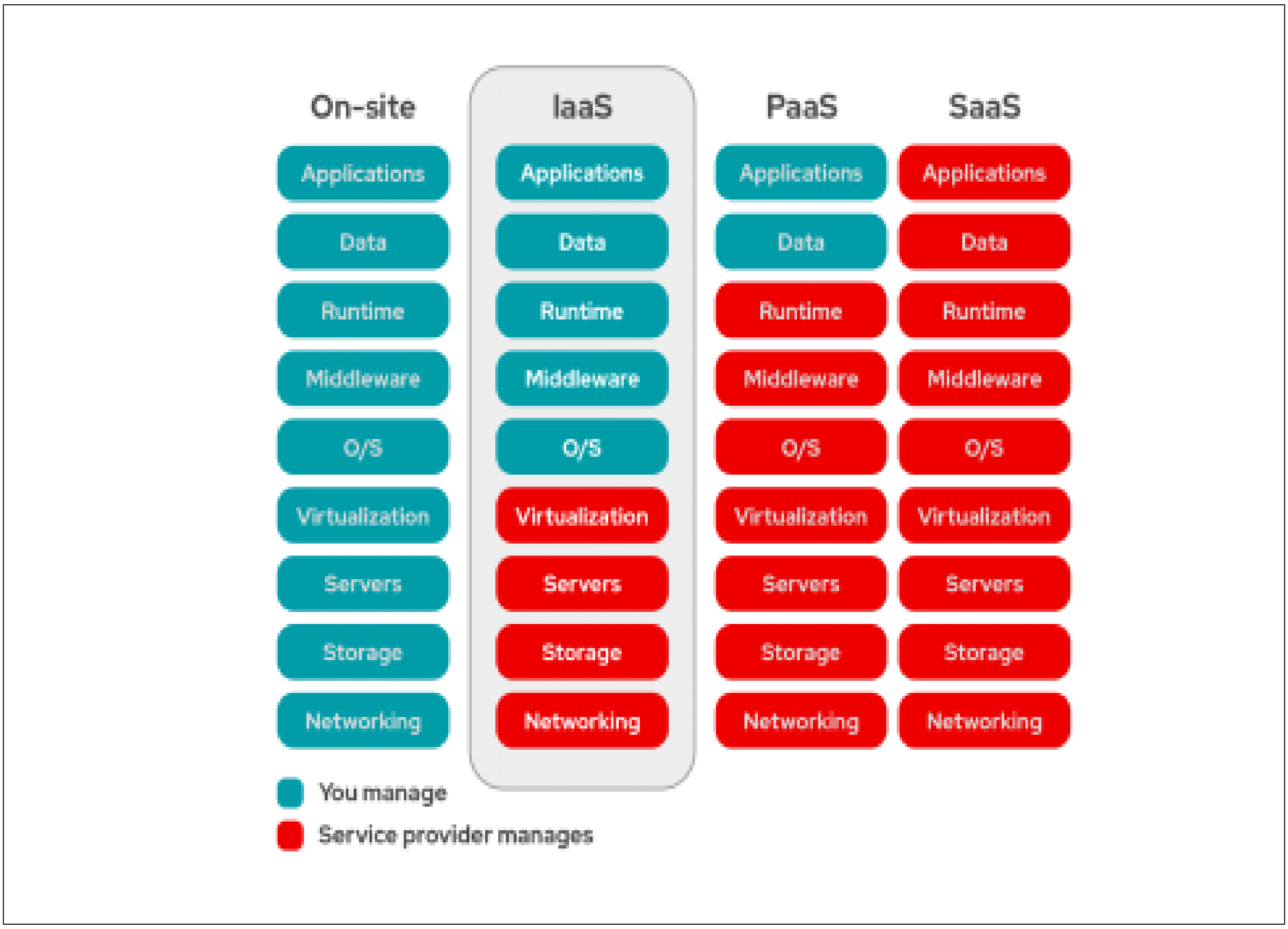
Cloud services

- Cloud services are infrastructure, platforms, or software that are hosted by third-party providers and made available to users through the internet.
- There are three main types of as-a-Service solutions: IaaS, PaaS, and SaaS. Each facilitates the flow of user data from front-end clients through the internet, to the cloud service provider's systems, and back—but vary by what's provided.

IaaS

- IaaS means a cloud service provider manages the infrastructure for you—the actual servers, network, virtualization, and data storage—through an internet connection.
- The user has access through an API or dashboard, and essentially rents the infrastructure.
- The user manages things like the operating system, apps, and middleware while the provider takes care of any hardware, networking, hard drives, data storage, and servers; and has the

responsibility of taking care of outages, repairs, and hardware issues. This is the typical deployment model of cloud storage providers.



PaaS

- PaaS means the hardware and an application-software platform are provided and managed by an outside cloud service provider, but the user handles the apps running on top of the platform and the data the app relies on.
- Primarily for developers and programmers, PaaS gives users a shared cloud platform for application development and management (an important DevOps component) without having to build and maintain the infrastructure usually associated with the process.

SaaS

- SaaS is a service that delivers a software application—which the cloud service provider manages—to its users.
- Typically, SaaS apps are web applications or mobile apps that users can access via a web browser. Software updates, bug fixes, and other general software maintenance are taken care of for the user, and they connect to the cloud applications via a dashboard or API.

- SaaS also eliminates the need to have an app installed locally on each individual user's computer, allowing greater methods of group or team access to the software.

Common cloud computing questions

1.Which cloud should I use?

That depends on what you're doing.

- Workloads with high volume or fluctuating demands might be better suited for a public cloud.
- Workloads with predictable use patterns might be better off in a private cloud.
- Hybrid clouds are the catch-all, because any workload can be hosted anywhere.

2.Which cloud is safest?

- Public clouds tend to have a wider variety of security threats due to multi-tenancy and numerous access points. Public clouds often split security responsibilities. For instance, infrastructural security can be the provider's responsibility while workload security can be the tenant's responsibility.
- Private clouds are thought to be more secure because workloads usually run behind the user's firewall, but that all depends on how strong your own security is.
- Hybrid cloud security is made up of the best features of every environment, where users and admins can minimize data exposure by moving workloads and data across environments based on compliance, audit, policy, or security requirements.

3.Which cloud costs more?

- You usually pay for what you use in a public cloud, though some public clouds (like the Massachusetts Open Cloud) don't charge tenants.
- Whoever set up a private cloud is usually responsible for purchasing or renting new hardware and resources to scale up.
- Hybrid clouds can include any on-prem, off-prem, or provider's cloud to create a custom environment that suits your cost requirements.

4.Which cloud has the best resources?

That depends on how you want to spend money. Do you want to incur capital expenses (CapEx) or operating expenses (OpEx)? This is the classic *scale-up* vs. *scale-out* question.

- Public cloud users seem to have unlimited access to resources, but accessing those resources is usually an operational expense.
- Deploying more private cloud resources requires buying or renting more hardware—all capital expenses.

- Hybrid clouds give you the option of using operating expenses to scale out or capital expenses to scale up.

4.4:Scope of cloud computing in Nepal

It is a glimpse of the obvious that the development of cloud computing requires internet access. You will know far better than I do the current limitations on internet access in Nepal. But it is getting better.

All of us in the favored West are full of admiration for the way in which Nepal has coped with such severe earthquakes and floods. It seems that internet access in Nepal may be wireless rather than fixed line. This will slow things in the short run but may confer an advantage in the long run as innovative technologies advance rapidly.

As soon as internet access is good, cloud computing in Nepal will, It is going to very important but not unlimited. **Some thoughts are below.**

There will be more market convergence around Google, AWS, and Microsoft.This is as easy to call as the sunset. Enterprises moving to public clouds are largely going with these three companies. Moreover, the big three are placing their technology bets in cloud computing right now, and this will continue for the foreseeable future.

Many thought this would be race of a dozen or so public cloud providers, but the market has taken care of most of them. Indeed, smaller cloud providers, such as Rack space, could not keep up with the investment and have moved to a narrower and more niche-oriented focus.

Anybody who is not AWS, Google, and Microsoft will have to grab a smaller piece of the market and hold on for dear life. Count on a few to move to the managed services space, such as Rack space, while others will provide a specific area of public cloud services, such as “bare metal” clouds. Still others will focus on specific verticals such as health care or finance.

Enterprises will become more aware of total cost of ownership, and this knowledge will slow or stop cloud adoption within some enterprises.

The value drivers presented earlier above are well understood and have a few years of proof in many enterprises. However, we also know cloud computing may not be cost effective or a good fit. That’s typically not seen until after the first few projects have been deployed, when the cost and benefit issues are better understood.

We’ll see the continued fall of the private cloud-only implementation.

Public clouds will be paired with private clouds to form hybrid or multiclouds, which provide enterprises with more cost efficiency and scalability. The number of enterprises employing only private clouds will fall substantially in the next few years.

New public cloud providers will have to quickly find a niche.

The market is dominated by existing players who can spend billions on both R&D and marketing. Startups are leveraging new and emerging cloud technology, such as containers, as a path for incremental growth. We'll likely see the same pattern around data analytics services, data storage services, Internet of things (IoT), and other more recent technology trends.

4.5:IOT & its features

Introduction of Internet Of Thing (IOT)

- The Internet of Things is an additional layer of information, interactions, transactions and actions that are added to the Internet through devices, equipped with capabilities for detection, analysis and communication of data, using protocols Internet.
- The IoT wants to connect all potential objects to interact with each other on the internet to provide a safe and comfortable life for humans. The Internet of Things (IoT) makes our world as connected as possible. Nowadays, we have an Internet infrastructure almost everywhere and we can use it anytime.
- Some real world examples of IoT are wearable fitness and trackers (like Fit bits) and IoT healthcare applications, voice assistants (Siri and Alexa), smart cars (Tesla), and smart appliances (iRobot).
- **Internet of Things (IoT)** is the networking of physical objects that contain electronics embedded within their architecture in order to communicate and sense interactions amongst each other or with respect to the external environment. In the upcoming years, IoT-based technology will offer advanced levels of services and practically change the way people lead their daily lives. Advancements in medicine, power, gene therapies, agriculture, smart cities, and smart homes are just a very few of the categorical examples where IoT is strongly established.
- ***IoT is network of interconnected computing devices which are embedded in everyday objects, enabling them to send and receive data.***
- Over 9 billion 'Things' (physical objects) are currently connected to the Internet, as of now. In the near future, this number is expected to rise to a whopping 20 billion.

Main components used in IoT:

1.Low-power embedded systems: Less battery consumption, high performance are the inverse factors that play a significant role during the design of electronic systems.

2.Sensors : Sensors are the major part of any IoT applications. It is a physical device that measures and detect certain physical quantity and convert it into signal which can be provide as an input to processing or control unit for analysis purpose.

Different types of Sensors :

a. Temperature Sensors
b. Image Sensors
c. Gyro Sensors
d. Obstacle Sensors
e. RF Sensor
f. IR Sensor
g. MQ-02/05 Gas Sensor
h. LDR Sensor
i. Ultrasonic Distance Sensor

3.Control Units : It is a unit of small computer on a single integrated circuit containing microprocessor or processing core, memory and programmable input/output devices/peripherals. It is responsible for major processing work of IoT devices and all logical operations are carried out here.

4.Cloud computing: Data collected through IoT devices is massive and this data has to be stored on a reliable storage server. This is where cloud computing comes into play. The data is processed and learned, giving more room for us to discover where things like electrical faults/errors are within the system.

5.Availability of big data: We know that IoT relies heavily on sensors, especially in real-time. As these electronic devices spread throughout every field, their usage is going to trigger a massive flux of big data.

6.Networking connection: In order to communicate, internet connectivity is a must where each physical object is represented by an IP address. However, there are only a limited number of addresses available according to the IP naming. Due to the growing number of devices, this naming system will not be feasible anymore. Therefore, researchers are looking for another alternative naming system to represent each physical object.

Characteristics of IoT:

a. Massively scalable and efficient
b. IP-based addressing will no longer be suitable in the upcoming future.
c. An abundance of physical objects is present that do not use IP, so IoT is made possible.
d. Devices typically consume less power. When not in use, they should be automatically programmed to sleep.

e. A device that is connected to another device right now may not be connected in another instant of time.
f. Intermittent connectivity – IoT devices aren't always connected. In order to save bandwidth and battery consumption, devices will be powered off periodically when not in use. Otherwise, connections might turn unreliable and thus prove to be inefficient.

Some benefits of using IoT are listed below:

1. More amount of data tends to better decision making. Like suppose sensors are implemented so they collect a large amount of data from different in different conditions which helps it to make good decisions in any new conditions and circumstances.
2. You can track and control any of your belongings and goods.
3. Used in the healthcare management system. Like monitoring and controlling the heartbeats and blood pressure of the patient.
4. Used in building smart homes.
5. Used in wearables. Like in smartwatch as GPS, heart rate monitoring, etc systems.

   Similarly, like them, there are lots of uses and benefits of IoT which are used to create automation and increase efficiency without human intervention.

Features of Internet of Thing (IOT)

1. Connectivity : IoT devices can be connected over Radio waves, Bluetooth, Wi-Fi, Li-Fi, etc. We can leverage various protocols of internet connectivity layers in order to maximize efficiency and establish generic connectivity across IoT ecosystems and industries.

2. Sensing : IoT uses Electrochemical, gyroscope, pressure, light sensors, GPS, Electrochemical, pressure, RFID, etc. to gather data based on a particular problem. For example for automotive use cases, IoT uses Light detection sensors along with pressure, velocity and imagery sensors.

3. Active Engagements : IoT device connects various products, cross-platform technologies and services work together by establishing an active engagement between them. IoT uses cloud computing in block chain to establish active engagements among IoT components.

4. Scale : IoT devices are being designed in such a way that they can be scaled up or down easily on demand.

5. Intelligence : In almost every IoT use case in today's world, the data is used to make important business insights and drive important business decisions.

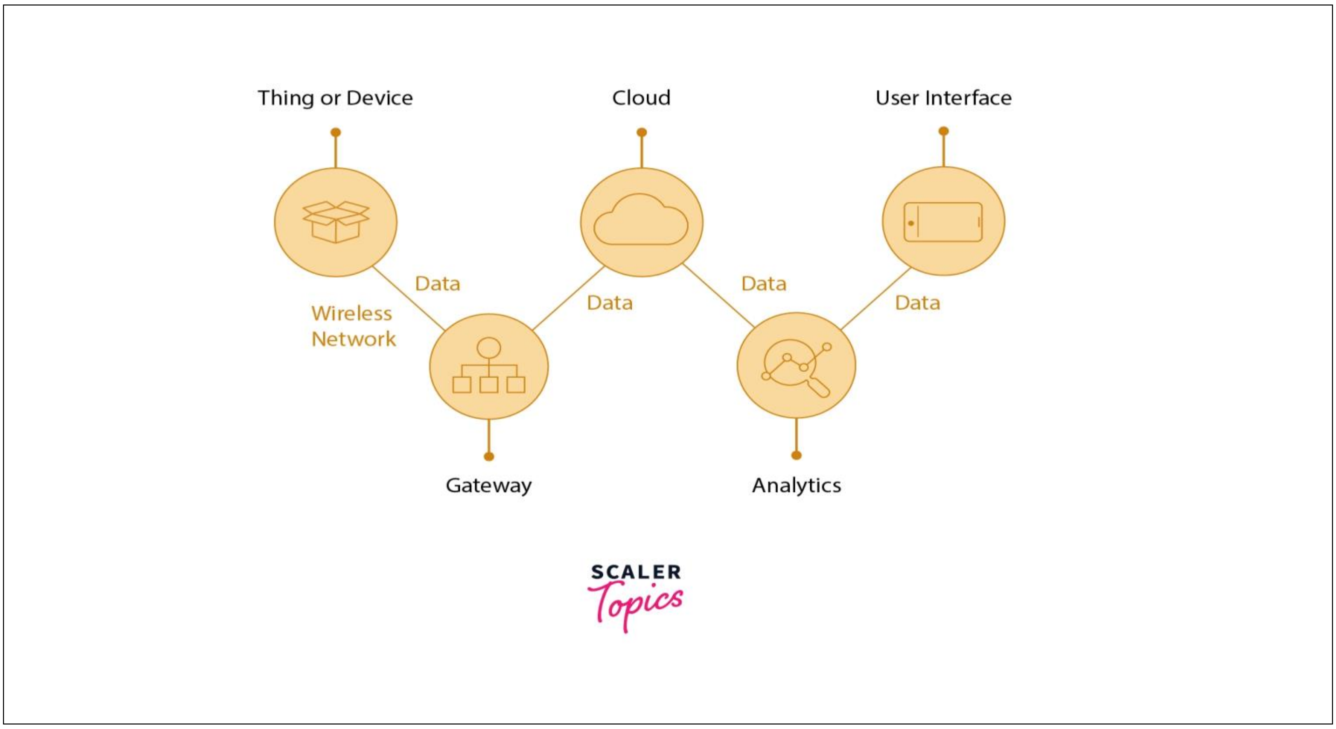
6.Energy: From end components to connectivity and analytics layers, the whole ecosystems demand a lot of energy.

7.Safety : One of the main features of the IoT ecosystem is security. In the whole flow of an IoT ecosystem, sensitive information is passed from endpoints to the analytics layer via connectivity components.

8.Integration : IoT integrates various cross-domain models to enrich user experience. It also ensures a proper trade-off between infrastructure and operational costs.

4.6 : IOT Components

Components of IOT



Things or Devices

The key physical items being tracked are Things or Devices. Smart sensors are connected to things/devices which further continues to collect data from the device and send it to the next layer, which is the portal or also called as the gateway Small smart sensors for a variety of applications are now possible because of new advancements in microelectronics.

Some commonly used sensors are:

a. Temperature sensors and thermostats
b. Pressure sensors
c. Humidity / Moisture level
d. Light intensity detectors
e. Moisture sensors
f. Proximity detection
g. RFID tags

User Interface

User interface also termed as UI is nothing but a user-facing program that allows the user to monitor and manipulate data.The user interface (UI) is the visible, tangible portion of the IoT device that people can interact with. Developers must provide a well-designed user interface that requires the least amount of effort from users and promotes additional interactions.

Cloud

Cloud storage is used to store the data which has been collected from different devices or things. Cloud computing is simply a set of connected servers that operate continuously(24*7) over the Internet.IoT devices, applications, and users generate massive amounts of data, which must be managed efficiently. Data collection, processing, management, and archiving are among the responsibilities of IoT clouds. The data can be accessed remotely by industries and services, allowing them to take critical decisions at any time.In the simplest terms, an IoT cloud is a network of servers optimized to handle data at high speeds for a large number of different devices, manage traffic, and analyze data with great accuracy. An IoT cloud would not be complete without a distributed management database system.

Analytics

After receiving the data in the cloud, that data is processed. Data is analyzed here with the help of various algorithms like machine learning and all.Analytics is the conversion of analog information via connected sensors and devices into actionable insights that can be processed, interpreted, and analyzed in depth. Analysis of raw data or information for further processing is a prerequisite for the monitoring and enhancement of the Internet of things (IoT).Among the most significant benefits of a well-designed IoT system is real-time smart analysis, which enables designers to spot anomalies in gathering information and respond quickly to avoid an undesirable situation. If information is collected correctly and at the right moment, network operators can plan for the next steps.

Network Interconnection

Over the past few years, the IoT has seen massive growth in devices controlled by the internet and connected to it. Although IoT devices have a wide variety of uses, there are some common things among them also along with the differences between them.

IoT is enabled by a variety of technologies. The network used to communicate with other devices in an IoT deployment is critical to the field, a position that numerous wireless or wired technologies can fill.

System Security

Security is a crucial component of IoT implementation, but this security point of view is too often overlooked during the design process. Day after day weaknesses within IoT are being attacked with evil intent – however, the majority of them that can be easily and inexpensively addressed.

A secure network begins with the elimination of weaknesses within IoT devices as well as the provision of tools to withstand, recognize, and recoup from harmful attacks.

Central Control Hardware

The two or more data flow among multiple channels and interfaces is managed by a Control Panel. The additional duty of a control panel is to convert various wireless interfaces and ensure that linked sensors and devices are accessible.

4.7 : Types of IOT wireless Networks

Types of IoT Wireless Networks

The Internet of Things (IoT) involves different kinds of continuously evolving technology, allowing for many connectivity options for connected device manufacturers. Each of these options has a range of advantages and disadvantages that must be weighed together.

What is Wireless Technology?

Wireless technology is a method of connection within an IoT system that includes sensors, platforms, routers, applications, and other systems. Each option has trade-offs between power consumption, bandwidth, and range. At a high level, there are standard wireless options like cellular (3G, 4G, 5G) and WiFi, and there are long-range options like LoRaWAN and LPWAN. Connected device companies should understand the basics of the top IoT technologies to determine the best wireless technology option for their needs.

5 Types of IoT Wireless Technologies and Their Use Cases

Before we get into how to decide which wireless technology is right for your business, let's look at the main options available today. These will meet the requirements of most use cases.

1. Cellular

One of the best-known types of IoT wireless technology is cellular, particularly in the consumer mobile market. Cellular networks provide reliable broadband communication that supports everything from streaming applications to voice calls. Since cellular is so well established, it offers very high bandwidth. Carriers offer Cat-M1 and NB-IoT, cellular options designed to compete with novel LPWAN technologies.Cellular works well in low power environments, especially with something like Cat-M1. And for non-mobile applications where power isn't a factor, cellular is also an excellent choice. With the introduction of 5G's ultra-low latency, wireless can effectively support the needs of things like time-sensitive industrial automation, real-time delivery of medical data, and public safety surveillance video. It will be exciting to watch this unfold.

2. Bluetooth and BLE (Bluetooth Low Energy)

Another well-known wireless technology in consumer circles is Bluetooth. This wireless personal area network (WPAN) is a short-range communication technology with optimization for power consumption (Bluetooth Low Energy) positioned to support small-scale consumer IoT applications.

Bluetooth and BLE are used for everything from fitness and medical wearables like smartwatches to smart home devices like home security systems, where data is communicated to smartphones. They work quite effectively with very short-range communications.

3. WiFi

WiFi has played a critical role in providing high-throughput data transfer in homes and for enterprises — it's another well-known IoT wireless technology. It can be quite effective in the right situations, though it has significant limitations with scalability, coverage, and high power consumption. Due to newer enterprise security practices, IoT devices are discouraged from being added to the same primary Wi-Fi networks as traditional employee computers or phones. In Zipit's experience, this has led more OEMs to leverage other technologies like cellular to both mitigate these security concerns as well as speed up deployment efforts.

The high energy requirements often make WiFi a poor solution for large networks with battery-operated sensors, such as smart buildings and industrial use. Instead, it's more effective with devices like smart home appliances. The latest WiFi technology, WiFi 6, does offer improved bandwidth and speed, though it's still behind other available options. And it carries security risks that other options don't.

4. LPWAN (Cat-M1/NB-IoT)

Low power wide area networks (LPWAN) provide long-range communication using small, inexpensive batteries. This family of technologies is ideal for supporting large-scale IoT networks where a significant range is required. However, LPWANs can only send small blocks of data at a low rate.

LPWANs are ideally suited for use cases that don't require time sensitivity or high bandwidth, like a water meter for example. They can be quite effective for asset tracking in a manufacturing facility, facility management, and environmental monitoring. Keep in mind that standardization is important to ensure the network's security, interoperability, and reliability. Although newer LPWAN is gaining popularity.

5. LoRaWAN

LoRaWAN is a powerful and emerging technology. It's similar to Bluetooth, but it offers a longer range for small data packets with low power consumption. LoRaWAN manages the communication frequencies, power, and data rate for all connected devices. So, LoRaWAN sensors communicate to a cellular gateway to send data to the cloud. It does require a back-haul transport, and partners like Zipit can provide the necessary cellular support.

The use cases for LoRaWAN are similar to those for LPWAN where there aren't high bandwidth requirements or time sensitivity for the transfer of data. But LoRaWAN differs in several ways. It's is a point-to-point wireless connection that does not have direct connection to the Internet, whereas LPWAN is a direct-to-internet connection. LoRaWAN requires a gateway to packet forward the data to the final destination in the cloud, and that gateway is typically a LoRa to Cellular gateway.

Since the range is high for LoRaWAN, sensors can be widely deployed over a large area. These sensors are simple and designed for small data packets that transmit infrequently, so it's well-suited for irrigation management, leak detection, logistics and transportation management, and asset or equipment tracking. Keep your eyes on LoRaWAN as it continues to evolve.

Unit-5:Artificial Intelligence

Contents

5.1: AI Introduction

What is Artificial Intelligence?

In today's world, technology is growing very fast, and we are getting in touch with different new technologies day by day.Here, one of the booming technologies of computer science is Artificial Intelligence which is ready to create a new revolution in the world by making intelligent machines.The Artificial Intelligence is now all around us. It is currently working with a variety of subfields, ranging from general to specific, such as self-driving cars, playing chess, proving theorems, playing music, Painting, etc.AI is one of the fascinating and universal fields of Computer science which has a great scope in future. AI holds a tendency to cause a machine to work as a human.

Artificial Intelligence is composed of two words Artificial and Intelligence, where Artificial defines "man-made," and intelligence defines "thinking power", hence AI means "a man-made thinking power."

So, we can define AI as:

"It is a branch of computer science by which we can create intelligent machines which can behave like a human, think like humans, and able to make decisions."

Artificial Intelligence exists when a machine can have human based skills such as learning, reasoning, and solving problems.With Artificial Intelligence you do not need to preprogram a machine to do some work, despite that you can create a machine with programmed algorithms which can work with own intelligence, and that is the awesomeness of AI.It is believed that AI is not a new technology, and some people says that as per Greek myth, there were Mechanical men in early days which can work and behave like humans.

Why Artificial Intelligence?

Before Learning about Artificial Intelligence, we should know that what is the importance of AI and why should we learn it. Following are some main reasons to learn about AI:

- With the help of AI, you can create such software or devices which can solve real-world problems very easily and with accuracy such as health issues, marketing, traffic issues, etc.
- With the help of AI, you can create your personal virtual Assistant, such as Cortana, Google Assistant, Siri, etc.
- With the help of AI, you can build such Robots which can work in an environment where survival of humans can be at risk.
- AI opens a path for other new technologies, new devices, and new Opportunities.

Goals of Artificial Intelligence

Following are the main goals of Artificial Intelligence:

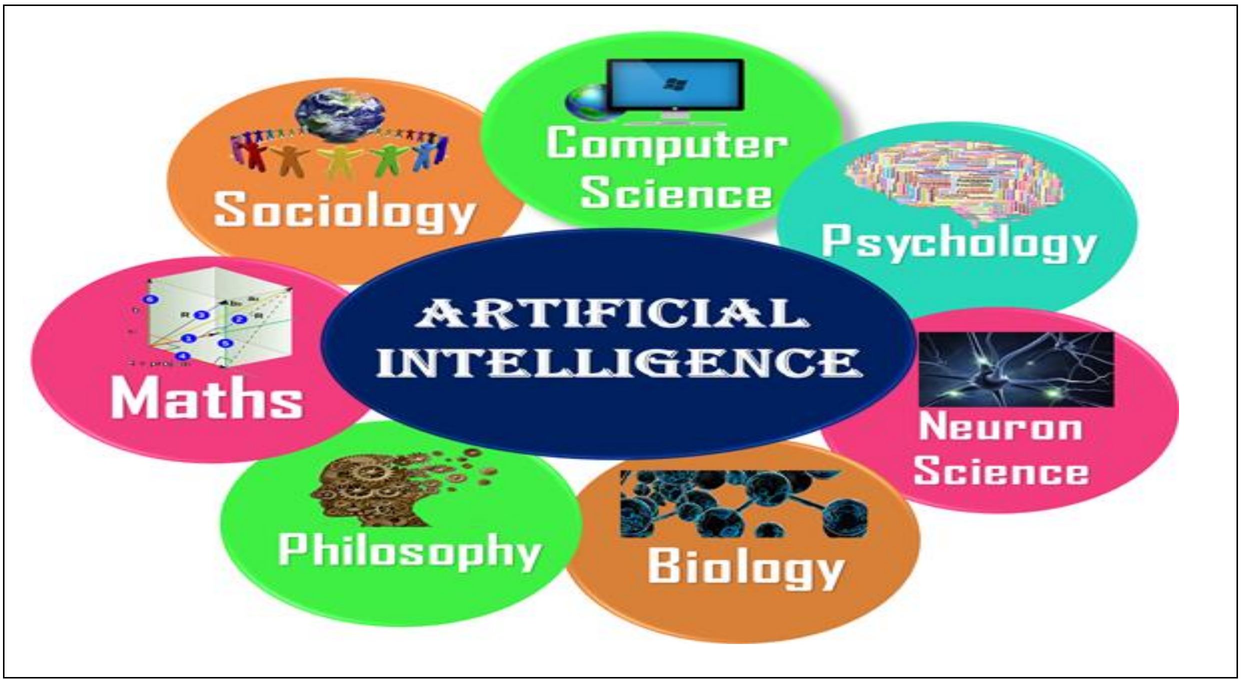
1. Replicate human intelligence
2. Solve Knowledge-intensive tasks
3. An intelligent connection of perception and action
4. Building a machine which can perform tasks that requires human intelligence such as:
    - Proving a theorem
    - Playing chess
    - Plan some surgical operation
    - Driving a car in traffic
5. Creating some system which can exhibit intelligent behavior, learn new things by itself, demonstrate, explain, and can advise to its user.

What Comprises to Artificial Intelligence?

Artificial Intelligence is not just a part of computer science even it's so vast and requires lots of other factors which can contribute to it. To create the AI first we should know that how intelligence is composed, so the Intelligence is an intangible part of our brain which is a combination of **Reasoning, learning, problem-solving perception, language understanding, etc**.

To achieve the above factors for a machine or software Artificial Intelligence requires the following discipline:

- Mathematics
- Biology
- Psychology
- Sociology
- Computer Science
- Neurons Study
- Statistics



Advantages of Artificial Intelligence

Following are some main advantages of Artificial Intelligence:

a. **High Accuracy with less errors:** AI machines or systems are prone to less errors and high accuracy as it takes decisions as per pre-experience or information.

b. **High-Speed:** AI systems can be of very high-speed and fast-decision making, because of that AI systems can beat a chess champion in the Chess game.

c. **High reliability:** AI machines are highly reliable and can perform the same action multiple times with high accuracy.

d. **Useful for risky areas:** AI machines can be helpful in situations such as defusing a bomb, exploring the ocean floor, where to employ a human can be risky.

e. **Digital Assistant:** AI can be very useful to provide digital assistant to the users such as AI technology is currently used by various E-commerce websites to show the products as per customer requirement.

f. **Useful as a public utility:** AI can be very useful for public utilities such as a self-driving car which can make our journey safer and hassle-free, facial recognition for security purpose, Natural language processing to communicate with the human in human-language, etc.

Disadvantages of Artificial Intelligence

Every technology has some disadvantages, and the same goes for Artificial intelligence. Being so advantageous technology still, it has some disadvantages which we need to keep in our mind while creating an AI system. Following are the disadvantages of AI:

a. **High Cost:** The hardware and software requirement of AI is very costly as it requires lots of maintenance to meet current world requirements.

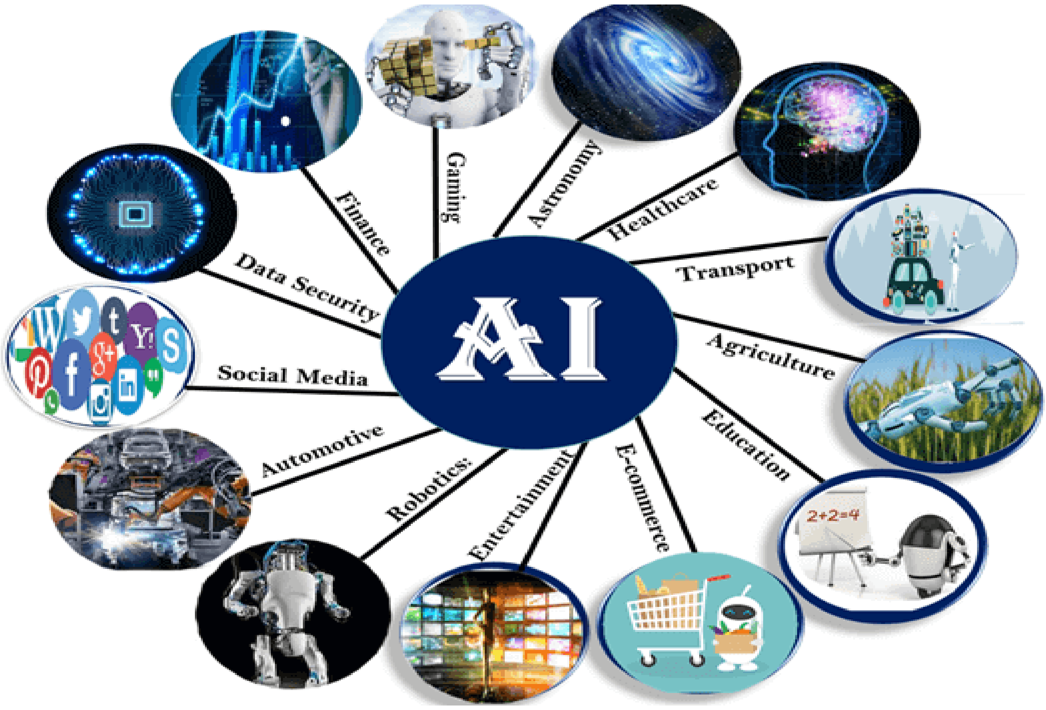
b. **Can't think out of the box:** Even we are making smarter machines with AI, but still they cannot work out of the box, as the robot will only do that work for which they are trained, or programmed.

c. **No feelings and emotions:** AI machines can be an outstanding performer, but still it does not have the feeling so it cannot make any kind of emotional attachment with human, and may sometime be harmful for users if the proper care is not taken.

d. **Increase dependency on machines:** With the increment of technology, people are getting more dependent on devices and hence they are losing their mental capabilities.

e. **No Original Creativity:** As humans are so creative and can imagine some new ideas but still AI machines cannot beat this power of human intelligence and cannot be creative and imaginative.

5.4:Application of Artificial Intelligence



Artificial Intelligence has various applications in today's society. It is becoming essential for today's time because it can solve complex problems with an efficient way in multiple industries, such as Healthcare, entertainment, finance, education, etc. AI is making our daily life more comfortable and fast.

Following are some sectors which have the application of Artificial Intelligence:

1. AI in Astronomy

Artificial Intelligence can be very useful to solve complex universe problems. AI technology can be helpful for understanding the universe such as how it works, origin, etc.

2. AI in Healthcare

In the last, five to ten years, AI becoming more advantageous for the healthcare industry and going to have a significant impact on this industry.

Healthcare Industries are applying AI to make a better and faster diagnosis than humans. AI can help doctors with diagnoses and can inform when patients are worsening so that medical help can reach to the patient before hospitalization.

3. AI in Gaming

AI can be used for gaming purpose. The AI machines can play strategic games like chess, where the machine needs to think of a large number of possible places.

4. AI in Finance

AI and finance industries are the best matches for each other. The finance industry is implementing automation, chatbot, adaptive intelligence, algorithm trading, and machine learning into financial processes.

5. AI in Data Security

The security of data is crucial for every company and cyber-attacks are growing very rapidly in the digital world. AI can be used to make your data more safe and secure. Some examples such as AEG bot, AI2 Platform,are used to determine software bug and cyber-attacks in a better way.

6. AI in Social Media

Social Media sites such as Facebook, Twitter, and Snapchat contain billions of user profiles, which need to be stored and managed in a very efficient way. AI can organize and manage massive amounts of data. AI can analyze lots of data to identify the latest trends, hashtag, and requirement of different users.

7. AI in Travel & Transport

AI is becoming highly demanding for travel industries. AI is capable of doing various travel related works such as from making travel arrangement to suggesting the hotels, flights, and best routes to the customers. Travel industries are using AI-powered chatbots which can make human-like interaction with customers for better and fast response.

8. AI in Automotive Industry

Some Automotive industries are using AI to provide virtual assistant to their user for better performance. Such as Tesla has introduced TeslaBot, an intelligent virtual assistant.

Various Industries are currently working for developing self-driven cars which can make your journey more safe and secure.

9. AI in Robotics:

Artificial Intelligence has a remarkable role in Robotics. Usually, general robots are programmed such that they can perform some repetitive task, but with the help of AI, we can create intelligent robots which can perform tasks with their own experiences without pre-programmed.

Humanoid Robots are best examples for AI in robotics, recently the intelligent Humanoid robot named as Erica and Sophia has been developed which can talk and behave like humans.

10. AI in Entertainment

We are currently using some AI based applications in our daily life with some entertainment services such as Netflix or Amazon. With the help of ML/AI algorithms, these services show the recommendations for programs or shows.

11. AI in Agriculture

Agriculture is an area which requires various resources, labor, money, and time for best result. Now a day's agriculture is becoming digital, and AI is emerging in this field. Agriculture is applying AI as agriculture robotics, solid and crop monitoring, predictive analysis. AI in agriculture can be very helpful for farmers.

12. AI in E-commerce

AI is providing a competitive edge to the e-commerce industry, and it is becoming more demanding in the e-commerce business. AI is helping shoppers to discover associated products with recommended size, color, or even brand.

13. AI in education:

AI can automate grading so that the tutor can have more time to teach. AI chatbot can communicate with students as a teaching assistant.

AI in the future can be work as a personal virtual tutor for students, which will be accessible easily at any time and any place.

Major Areas of Artificial Intelligence

All developments in AI are inspired by the nature. Generally any artifact or any theory is inspired by natural phenomena. Let's see some examples

Nature	Artifact
Human Face	Face of a Vehicle
Bird	Airplane
Fish	Submarine
Elephant	Backo

This can be applied to AI techniques as well. But we should not forget there are also man-made things such as perfume, drugs which are not in the nature. Some artificial things are better than natural things. Examples are rubber, engine-oil.

The major areas of AI are

1.Expert Systems — This is a model of human experts like doctors, lawyers, engineers, scientists, carpenters, musicians solve problems.

2. Artificial Neural Network (ANN) — ANN is inspired by how the brain is composed with large number of Neurons and Connections. Neurons work as processors and Connections work as memories.

3. Genetic Algorithm — Chromosomes and Genes give the inspiration to make the technology of Genetic Algorithm.

4. Fuzzy Logic — Human intelligence has a feature that is operating set of values in the proximity rather than operate on exact values. This nature is inspired for the Fuzzy Logic.

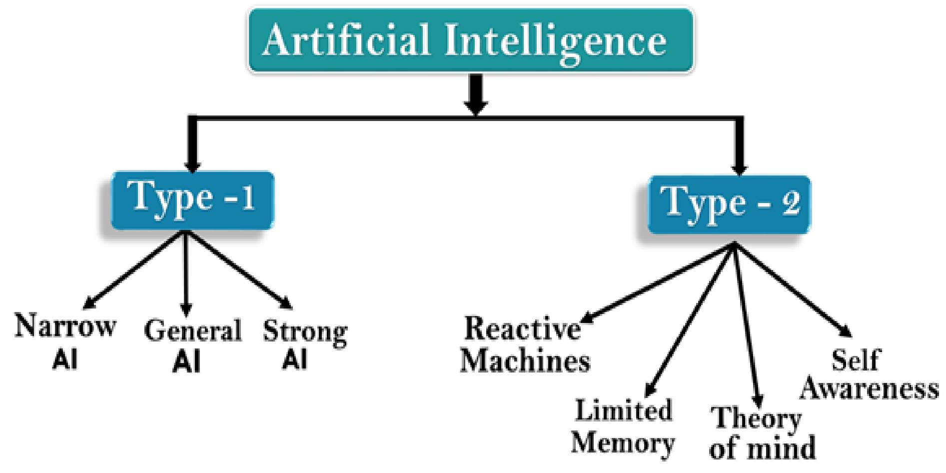
5. Robotics — This is a model of how human or animals do some physical tasks.

6. Natural Language Processing (NLP) — This is an area concern with natural language understanding and translation between natural languages.

7. Multi Agent Systems — These systems are based on problem solving by team work with the use of communication or message passing.

Types of Artificial Intelligence

Artificial Intelligence can be divided in various types, there are mainly two types of main categorization which are based on capabilities and based on functionally of AI. Following is flow diagram which explain the types of AI.



AI type-1: Based on Capabilities

1. Weak AI or Narrow AI:

- Narrow AI is a type of AI which is able to perform a dedicated task with intelligence.The most common and currently available AI is Narrow AI in the world of Artificial Intelligence.
- Narrow AI cannot perform beyond its field or limitations, as it is only trained for one specific task. Hence it is also termed as weak AI. Narrow AI can fail in unpredictable ways if it goes beyond its limits.

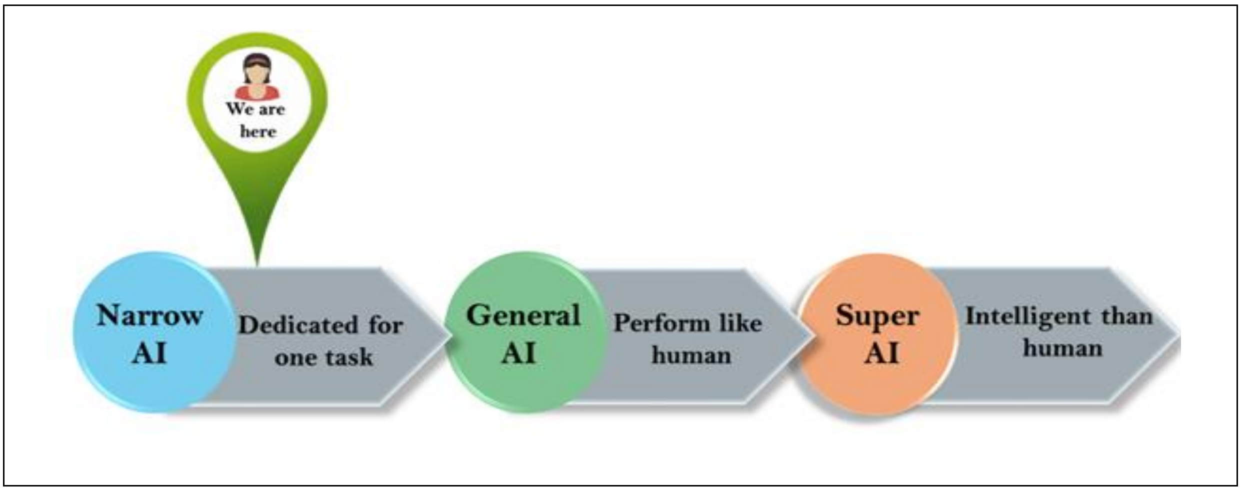
- Apple Siriis a good example of Narrow AI, but it operates with a limited pre-defined range of functions.
- IBM's Watson supercomputer also comes under Narrow AI, as it uses an Expert system approach combined with Machine learning and natural language processing.
- Some Examples of Narrow AI are playing chess, purchasing suggestions on e-commerce site, self-driving cars, speech recognition, and image recognition.

2. General AI:

- General AI is a type of intelligence which could perform any intellectual task with efficiency like a human.
- The idea behind the general AI to make such a system which could be smarter and think like a human by its own.
- Currently, there is no such system exist which could come under general AI and can perform any task as perfect as a human.
- The worldwide researchers are now focused on developing machines with General AI.
- As systems with general AI are still under research, and it will take lots of efforts and time to develop such systems.

3. Super AI:

- Super AI is a level of Intelligence of Systems at which machines could surpass human intelligence, and can perform any task better than human with cognitive properties. It is an outcome of general AI.
- Some key characteristics of strong AI include capability include the ability to think, to reason,solve the puzzle, make judgments, plan, learn, and communicate by its own.
- Super AI is still a hypothetical concept of Artificial Intelligence. Development of such systems in real is still world changing task.



Artificial Intelligence type-2: Based on functionality

1. Reactive Machines

- Purely reactive machines are the most basic types of Artificial Intelligence.
- Such AI systems do not store memories or past experiences for future actions.
- These machines only focus on current scenarios and react on it as per possible best action.
- IBM's Deep Blue system is an example of reactive machines.
- Google's AlphaGo is also an example of reactive machines.

2. Limited Memory

- Limited memory machines can store past experiences or some data for a short period of time.
- These machines can use stored data for a limited time period only.
- Self-driving cars are one of the best examples of Limited Memory systems. These cars can store recent speed of nearby cars, the distance of other cars, speed limit, and other information to navigate the road.

3. Theory of Mind

- Theory of Mind AI should understand the human emotions, people, beliefs, and be able to interact socially like humans.
- This type of AI machines are still not developed, but researchers are making lots of efforts and improvement for developing such AI machines.

4. Self-Awareness

- Self-awareness AI is the future of Artificial Intelligence. These machines will be super intelligent, and will have their own consciousness, sentiments, and self-awareness.
- These machines will be smarter than human mind.
- Self-Awareness AI does not exist in reality still and it is a hypothetical concept.

How AI transformed the world

The AI has become the characteristic features of the modernized and globalized world. AI is everywhere. It's omnipresent in our daily lives. Even when we don't see it working, it sees us, hears us, and is repeatedly learning from our behaviors. In Gmail, Whatsapp, or other social media platforms, we can see the use of AI predicting what we wanted to write about.

There is no exhaustive definition of AI. In the words of Stuart Russell and Peter Norvig, Artificial Intelligence is a system that thinks like humans, acts like humans, thinks rationally, and acts rationally.

In India, in 2019, Kerala police inducted a robot for police work. Then, it was the turn of Chennai to get its second robot-themed restaurant in 2019 itself. These developments symbolize the arrival of AI at the doorsteps of people. Even differently-abled citizens in India and Nepal are in a position to be benefitted from different mobile applications. They could get knowledge on a particular subject in their own language as Google has introduced live transcribe options for over 70 languages, including Nepali and Hindi. By using AI, we can scan a book, analyze and understand the texts in a simple way and then we can press command for creating questions based on a particular text. It will be beneficial for students in many ways. As such, neither Nepal nor India can afford to ignore AI.

AI in Nepal and India

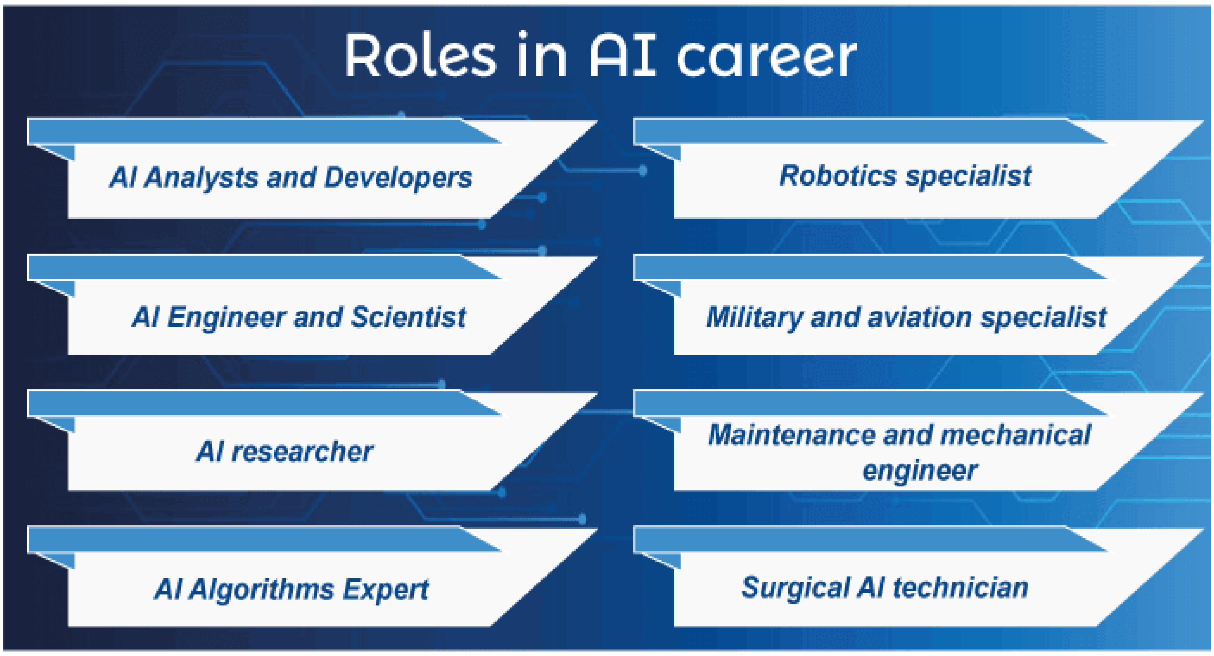
- Nepal lacks policy-level intervention on Artificial Intelligence but the use of AI-based technologies in banks and health sectors shows that Nepal is not far behind in technological advancements.
- The Naulo Restaurant in Kathmandu has five robot waiters under the slogan of "where the food meets technology". The robots are designed and manufactured by *Paaila* Technology, which is a Nepali company established by young engineers specializing in robotics and AI. Interestingly, Naulo is the first digitalised robotic restaurant in South Asia.
- Machine Learning (ML), a subfield of AI, is widely adopted by banks across the globe. The scope of AI is widening day by day with the advent of digital platforms and smartphones. Nepal's banking industries have digitization, fin-tech, mobile banking, SMS banking, internet banking, payment gateways and e-wallets which are also, in one or other way, use of AI.
- Previously, the use of AI could be seen in Nepal SBI Bank's robotics introduced in 2017 and Macchapuchhre Bank's MAYA, a chat assistant on its website, and

Facebook messenger in 2018. The use of AI in Banks and Financial Institutions and other sectors has proved to be a boon, for the software helps banks to link transactions, detect fraud/disasters and thereby ensure regulatory compliance.

- The government of Nepal has endorsed a program of "Digital Nepal Framework, 2019" which is to be implemented in five years, with the vision of 'digital Nepal for good governance, development and prosperity'. With this policy level intervention, the Himalayan Republic aims to ensure digital development in agriculture, education, health, energy, finance, tourism and urban infrastructure. The Framework is like a lamppost showing the vivid path to the country in its journey toward becoming a digital state.
- The Nagarik (citizen) App, released by the government in 2019 but implemented in January 2021, is an online service for various government and public bodies. This App provides information about government and public offices. It would have been better had the government launched an App like India's Aarogya Setu which is integrated with the vaccine registration and it has the potential to trace Covid-19 infected persons relatively easily. In fiscal year 2020-21, the government has allocated budget for increasing the quality and reliability of broadband internet service within two years.
- The AI has been making progress in Nepal and possibly changes will be seen at a time when the government and non-government sectors digitize their official business and set aside paperwork. Like other democracies, Nepal has a wealth of software talents in AI and IT sector. AI has been introduced as a discipline in university curricula. Kathmandu University has launched B Tech and M Tech programs in Artificial Intelligence from the academic session of 2021.

Scope of AI (AI Careers)

Fresher's should analyze their competencies and skills and choose a better AI role with the potential for upward mobility. The future scope of Artificial Intelligence continues to grow due to new job roles and advancements in the AI field. The various roles in an AI career are as follows:



a. AI Analysts and Developers
b. AI Engineer and Scientist
c. AI researcher
d. AI Algorithms Expert
e. Robotics specialist
f. Military and aviation specialist
g. Maintenance and mechanical engineer
h. Surgical AI technician

The future of AI

The future of Artificial Intelligence is bright in India, with many organizations opting for AI automation. It is essential to understand the recent developments in AI to find suitable job roles based on your competencies.

The scope of Artificial Intelligence is limited to domestic and commercial purposes as the medical and aviation sectors are also using AI to improve their services. If AI is outperforming human efforts, then opting for AI automation will reduce costs in the long run for a business.

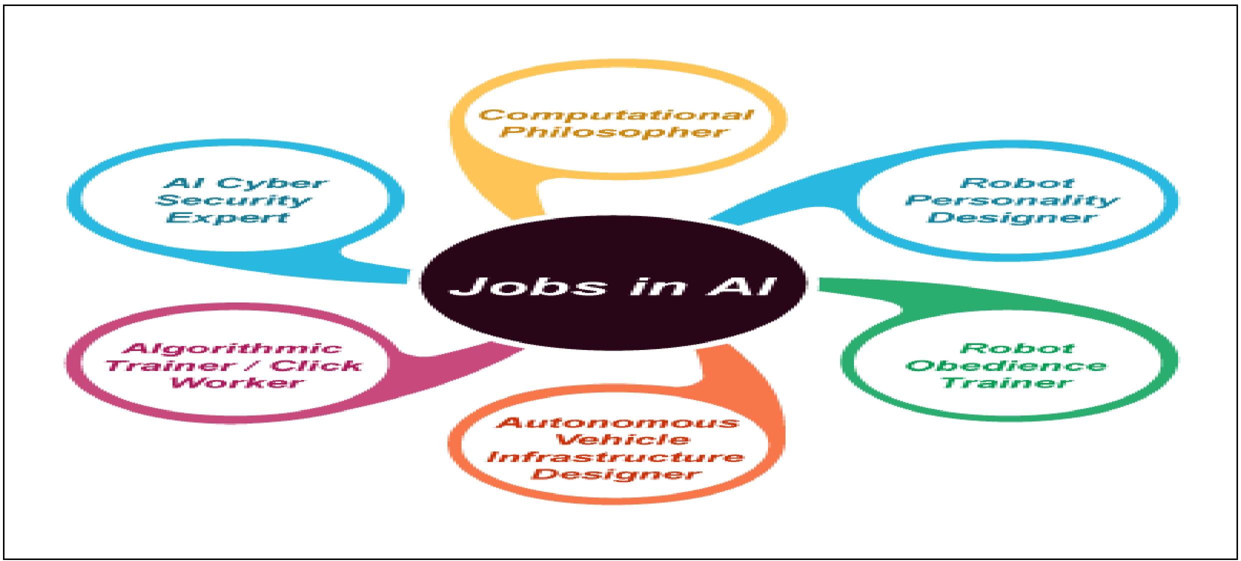
Automation in operational vehicles has created a buzz in the logistics industry as it is expected that automated **trucks/vehicles** may soon be used.

Due to the bright scope of Artificial Intelligence in the future, the number of AI start-ups is expected to increase in the coming years. Indicating the opportunities, the number of AI start-ups in India has increased significantly.

Moreover, India's talent gap for specialist AI developers is huge, and AI experts are needed by businesses more than ever. Businesses don't want to miss out on any technology that can revolutionize their business processes.

Jobs in AI

The name of the designation in the AI field may be different. Some of the top jobs in AI **(India - 2021**) are as follows:



Computational Philosopher - A computational philosopher concerned with teaching human ethics and values in AI algorithms. For example, if a robot is being developed for household chores, it should be designed to listen and follow orders from its employer.

Robot Personality Designer - A dedicated Robot Personality Designer designs the digital personality of a machine/robot.

Robot Obedience Trainer - A robotic obedience trainer teaches the machine/robot to follow instructions and obey obstacles. With more and more robots being introduced in homes, military strategies, etc., the future of Artificial Intelligence is bright.

Autonomous Vehicle Infrastructure Designer - An autonomous vehicle designer develops digital interfaces that help them work independently. The scope of a bright artificial intelligence future can fuel the development of autonomous vehicles in various industries.

Algorithmic Trainer / Click Worker - They work with AI algorithms and train them to recognize instructions, emotions, moods, images, speech, etc. They train AI algorithms to interact with their surroundings and take appropriate actions autonomously.

AI Cyber Security Expert - An AI Cyber Security expert develops algorithms that can identify the theft/risk associated with the system and take actions to eliminate it autonomously. As new types of cyber attacks evolve every day, AI is being used in cyber security to detect them. The future of Artificial Intelligence (**AI Cyber Security**) is also bright in the Asia Pacific region.

The future scope of Artificial Intelligence in India

The adoption of Artificial Intelligence in India is promising. However, it is currently in its early stages. While some industries, such as IT, manufacturing, automobiles, etc., are taking advantage of the prowess of AI, there are still many areas in which its potential has not been explored.

The immense potential present in AI can be understood by the various other technologies included under the umbrella of AI. Examples of such technologies include self-improvement algorithms, machine learning, pattern recognition, big data, and many others.

It is predicted that hardly any industry will be left untouched by this powerful tool in the next few years. It is the reason why AI has so much potential to grow in India.

In this comprehensive blog, we have discussed some of the areas in which AI is being used:

Artificial Intelligence job opportunities

According to a report published by Forbes, AI job opportunities are continuously increasing at **74%** annually. It is a no-brainer that today, AI is one of the most in-demand technologies, and it has an impact in almost every field. As a result, the demand for AI professionals continues to grow. As the number of job opportunities increases, it is the best time to explore your career in AI.

Below, we have compiled a list of different areas where AI is used or has immense potential to grow.

1: Banking

Banking is nothing new, thanks to the trends in Artificial Intelligence and Machine Learning technologies. The sector has rapidly adopted technology to keep up to date with the current market trends. It uses this technology to record customer data, which was previously a monotonous manual task.

With the rapid increase in the amount of data generated and stored in the banking sector today, Artificial Intelligence and ML allow professionals to do this accurately and efficiently.

How AI has made a significant difference in banking includes better **customer support, enhanced data quality, fraud prevention, digital assistants**, and more.

One of the most progressive sectors in the world today is healthcare. In the next section, you will read how Artificial Intelligence has affected this sector and how it will continue to do so.

2: Health care and medicine

According to one of the studies conducted by Forbes, the realm of AI can add value to life, as has already been observed over the years. The healthcare sector uses this technology to its advantage in several ways and constantly innovates.

AI use case is the collaborative Cancer Cloud, developed by Intel and the Knight Career Institute. Cloud collects past data of cancer patients and other patients with similar diseases to help doctors diagnose cancer early based on the symptoms they have shown and compare them with previously available data. The best treatment for this deadly disease is to prevent it from reaching its advanced stage.

In addition, Eve, an AI-based robot built by a team of scientists from the top universities of Aberystwyth, Manchester, and Cambridge, discovered an element often found in toothpaste that can cure malaria. It is proof that Artificial Intelligence will play an important role in the medical field in the coming times.

AAI is also used in health care and medicine in other similar fields, such as drug testing, synthetic biology, etc. You can also be sure that AI will accelerate the process of scientific research and development, which may well aid this field.

Contents

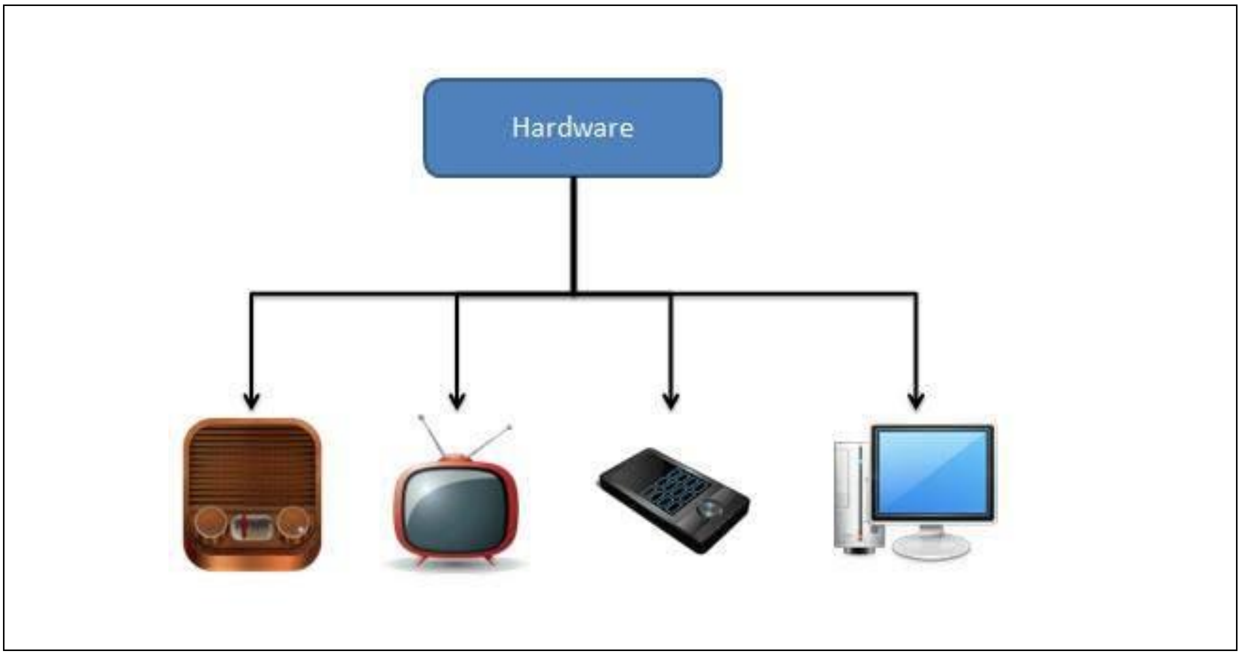
Multimedia Introduction

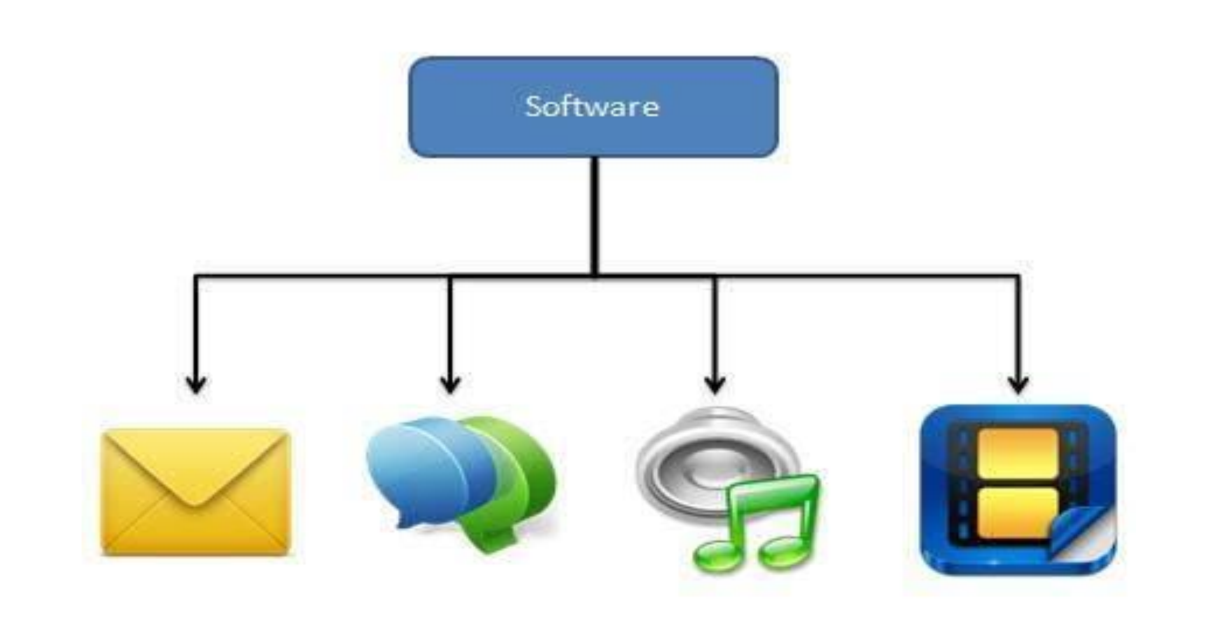
Multimedia is an interactive media and provides multiple ways to represent information to the user in a powerful manner. It provides an interaction between users and digital information. It is a medium of communication.Some of the sectors where multimedia is used extensively are education, training, reference material, business presentations, advertising and documentaries.

Definition of Multimedia

By definition Multimedia is a representation of information in an attractive and interactive manner with the use of a combination of text, audio, video, graphics and animation. In other words we can say that Multimedia is a computerized method of presenting information combining textual data, audio, visuals (video), graphics and animations. For examples: E-Mail, Yahoo Messenger, Video Conferencing, and Multimedia Message Service (MMS).

Multimedia as name suggests is the combination of Multi and Media that is many types of media (hardware/software) used for communication of information.





What is Multimedia?

Multimedia is an engaging kind of media that offers a variety of effective ways to convey information to users. Users can interact with digital information through it. It serves as a communication tool. Education, training, reference materials, corporate presentations, marketing, and documentary are a few industries that heavily utilize multimedia.Multimedia, by definition, is the use of text, audio, video, graphics, and animation to convey information in an engaging and dynamic way. In other terms, multimedia is a technological way of presenting information that combines audio, video, images, and animations with textual data. Examples include video conferencing, Yahoo Messenger, email, and the Multimedia Messaging Service (MMS Service (MMS).As the name implies, multimedia is the combination of the words "multi" and "media," which refers to the various media (hardware/software) utilized for information transmission.

Advantages and disadvantanges of multimedia

Some of the Advantages and Disadvantages of Multimedia are as follows. So let us check it out some information on advantages and disadvantages to know more about multimedia.It is a combination of different types of media which includes text, audio, video, graphics, animation etc.,

Advantages of Multimedia

1. **Increases learning effectiveness** : Multimedia uses images, audio, animations and other media which stimulates the brain and increases learning. Both attention and retention of student increases. Students can identify and solve various problems effectively using multimedia

2. **More appealing over traditional** :Multimedia content is more appealing, more attractive and able to engage more people (Audience or Reader). People prefer multimedia compared to traditional media as it contains rich information.
3. **Improves personal Communication** :Multimedia improves interpersonal communication. Multimedia increases communication efficiency, effectiveness and potention to communicate message or information.
4. **Reduces training costs** :Multimedia reduces the time it takes to learn and thereby it reduces cost. Multimedia can reduce the cost of training department in various ways.
5. **Easy to use** : Multimedia is making easier the things let they can easy understand. Content is easy to draft using different types of multimedia.
6. **Provides high Quality of Presentations** : Multimedia contains interactive videos, audio and other visual content which increases the quality of presentation. Students expect multimedia presentation while learning.
7. **User Friendly** : Multimedia improves user interface and it is easy to use. Information in text, image or audio form can be possible to promote information.
8. **Give information Cost** : Multimedia gives an opportunity to influence the presentation by adding information.
9. **Give information to individual** : Multimedia uses combination of different content and gives rich information to users. It uses infographic or audio and video to give detail information to individual.
10. **Multi Sensorial** : Multimedia is a multiple sensorial media, it makes to inclusion of layered sensory stimulation possible and interaction through various kinds of sensory channels.
11. **Integrated and Interative** :Multimedia is easy to integrate and makes it interactive.
12. **Entertaining and Educational** : Multimedia is not only used just for advertisement but also useful for education is schools as well as for entertainment.
13. **Creativity** : Multimedia creativity is very easy and effective if all kinds of media are integrated with each other.
14. **Cost Effective** : Multimedia is cost effective in online teaching, it reduces cost of training and other extra expenses.
15. **Wide variety of Support** : Through multimedia channels wide variety of support is available.
16. **Trendy** : Modern design and multiple media helps to create trendy multimedia presentations.

Disadvantages of Multimedia

1. **Misuse of Multimedia** : It becomes difficult for teachers to monitor students, some students may play games or surfing web pages and may not focus on their studies
2. **Lost in cyberspace** : People may waste their time on surfing unnecessary information on internet and may not focus on important issues.
3. **Non Interactive** : Multimedia may be less interactive. Teachers teaching students may be more interactive compared to videos or internet teaching students
4. **Text intensive content** : Lot of information is available on internet so it takes time find out the information which is needed.
5. **Time consuming** : Effective multimedia content takes time while creation.
6. **Too unrealistic** : Large files such as audio or video effects , the time multimedia takes to load your presentation.

What are the components of multimedia?

The following are typical multimedia elements:

1) Text - Text appears in all multi-media projects to some extent. To match the successful presentation of the multimedia program, the text may be presented in a variety of font styles and sizes.

2) Graphics - The multimedia program is appealing because of its graphics. People frequently find it difficult to read long passages of text on screens. As a result, visuals are frequently utilized instead of writing to convey ideas, give context, etc. Graphics can be of two different types:

Bitmap - Bitmap images are authentic pictures that can be taken using tools like digital cameras or scanners. Bitmap pictures are often not modifiable. Memory use for bitmap pictures is high.

Vector Graphics - Computers can draw vector graphics because they just need a little amount of memory. These images can be changed.

3) Animation - A static picture can be animated to appear to be in motion. A continuous succession of static images shown in order is all that makes up an animation. Effective attention-getting may be achieved by the animation. Additionally, animation adds levity and appeal to a presentation. In multimedia applications, the animation is fairly common.

4) Audio - Speech, music, and sound effects could all be necessary for a multimedia application. They are referred to as the audio or sound component of multimedia. Speaking is a fantastic educational tool. Analog and digital audio are both kinds. The initial sound signal is referred to as analog audio or sound. Digital sound is saved on a computer. Digital audio is therefore utilized for sound in multimedia applications.

5) Video - The term "video" describes a moving image that is supported by sound, such as a television image. A multimedia application's video component conveys a lot of information quickly. For displaying real-world items in multimedia applications, digital video is helpful. If uploaded to the internet, the video really does have the highest performance requirements for computer memory and bandwidth. The quality of digital video files may still be preserved while being saved on a computer, similarly to other data. A computer network allows for the transport of digital video files. The digital video snippets are simple to modify.

What are the applications of multimedia?

The typical areas where multimedia is applied are listed below.

1) For entertainment purposes - Multimedia marketing may significantly improve the promotion of new items. Both advertising and marketing staff had their doors opened by the economical communication boost provided by multimedia. Flying banner presentations, video transitions, animations, and audio effects are just a few of the components utilized to create a multimedia-based advertisement that appeals to the customer in a brand-new way and encourages the purchase of the goods.

2) For education purposes - There are currently a lot of educational computer games accessible. Take a look at an illustration of an educational app that plays children's rhymes. In addition to merely repeating rhymes, the youngster may create drawings, scale items up or down, and more. There are many more multimedia products on the market that provide children with a wealth of in-depth knowledge and playing options.

3) For business purposes - There are several commercial uses for multimedia. Multimedia and communication technologies have made it possible for information from international work groups. Today's team members can work remotely and for a variety of businesses. A global workplace will result from this. The following facilities should be supported by the multimedia network:

Office needs

Records management

Employee training

Electronic mail

Voice mail

4) For marketing purposes - Multimedia marketing may significantly improve the promotion of new items. Both advertising and promotion staff had their doors opened by the economical communication boost provided by multimedia. Flying banner presentations, video transitions, animations, and audio effects are just a few of the components utilized to create a multimedia-based advertisement that appeals to the customer in a brand-new way and encourages the purchase of the goods.

5) For banking purposes - Another public setting where multimedia is being used more and more recently is banks. People visit banks to open savings and current accounts, make deposits and withdrawals, learn about the bank's various financial plans, apply for loans, and other things. Each bank wants to notify its consumers with a wealth of information. It can employ multimedia in a variety of ways to do this. The bank also has a PC monitor in the clients' rest area that shows details about its numerous programs. Online and internet banking have grown in popularity recently. These heavily rely on multimedia. As a result, banks are using multimedia to better serve their clients and inform them of their appealing financing option.